

Generalized s-d model for Wannier-Mott excitons in layered magnetic semiconductors

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The recent discovery of excitons coupled to the magnetic order, and the consequent strong magneto-optic responses, in some van der Waals magnetic semiconductors has triggered intense activity at the interface of magnetism and semiconductor optics. Here, we present an analytically tractable minimal model that describes magnetic order, electrons, holes, and excitons within a unified framework, thereby capturing a wide range of phenomena. It treats the magnetic order and itinerant carriers to be comprised by distinct electronic orbitals that are mutually coupled via orbital-dependent onsite exchange, similar to the treatment of metallic magnets using an s-d model. Investigating CrSBr bilayer as a case study, we benchmark our model and its predictions against recent experimental and ab-initio results finding good agreement as well as new insights enabled by the model's simplicity. Examining the optical selection rules, we find the conservation of a quantum number formed from a combination of spin and layer pseudospin to be a useful guiding principle, even in noncollinear magnetic configurations. Our analysis finds a series of bright and dark excitonic states in such layered A-type antiferromagnets. The presented framework should be valuable in achieving intuitive understanding of recently discovered excitonic phenomena and guiding the discovery of other excitonic states in layered magnetic semiconductors.

I. INTRODUCTION

The recent discovery of layered van der Waals magnets has provided material platforms hosting a number of unique phenomena [1–3], such as two-dimensional magnetism in mono- and bilayers, with profound scientific and technological implications. A key ingredient that underlies this opportunity is the coexistence of strong crystalline intralayer interactions, which characterize their robust properties, and weak interlayer van der Waals coupling that admits a convenient external control. A special place in this class is held by the magnetic semiconductors, a prominent example being CrSBr [4–6], which are enabling unprecedented magneto-optic effects [4, 7, 8] due to their hosting robust bright excitons [9], bound electron-hole pairs, that couple strongly to the magnetic order. Since their recent discovery, these excitonic states have been investigated intensively, experimentally and using ab-initio methods, uncovering a wealth of phenomena [9]. Among these, we note their strong tunability via the magnetic order control as well as their different spatial extents associated with their variable Frenkel-like and Wannier-like characters [10–12]. While the early discoveries focused on optically bright excitons playing a role in the optical responses, recent research has gone beyond optics and also investigates excitonic transport [13, 14] as well as dark excitons [15–19].

Complementary to the optics, these magnetic semiconductors also harbor magnonic and spintronic phenomena, such as magnetoresistance effect [20], magnetic hysteresis [21–23], and magnon spin transport [24]. These effects

can be understood in terms of the magnetic order parameter, described via the Landau-Lifshitz framework, combined with the independent particle description of electrons and holes developed in semiconducting spintronics [25–27]. This class of effects is valuable for both probing the magnetic and crystalline order in the materials as well as for storing information magnetically [21–23].

In this article, we aim to unify these seemingly complementary phenomena and theoretical approaches into a simplified analytically tractable framework. We formulate a generalized s-d model to describe the magnetic order, electrons, holes, and excitons in layered A-type antiferromagnets with ferromagnetic intralayer exchange [9]. In developing this framework, we are inspired by the treatment of conventional metallic magnets via the s-d model that considers the magnetic order to be captured by the localized d electrons and electrical conduction to be contributed by the itinerant s electrons. Despite the actual metals often consisting of hybridized orbitals that do not allow this clear separation, the s-d model has been successful in analytically capturing and predicting a broad range of phenomena by incorporating an s-d exchange coupling treated as a phenomenological material parameter [28–30].

With the goal of accomplishing a minimal framework, we consider a tight-binding Hamiltonian with each site associated with a localized spin magnetic moment and two electronic orbitals that capture the conduction and valence bands. Their orbital-dependent onsite exchange coupling with the localized spin is found to be important for capturing the key material properties. Formulating our general model, we take up a detailed case study of CrSBr bilayers with the aim of benchmarking our approach against previous results [4, 18, 31–33] and devel-

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opening new insights enabled by our simple analytic model. We determine the phenomenological parameters, such as intersite hopping amplitudes and onsite exchange couplings, entering our model by comparing the previously established band structure with our evaluation. Our simple model enables formulation of optical selection rules thereby labeling bright and dark transitions in terms of a combined spin and layer-pseudospin quantum number. Investigating Wannier-Mott excitonic states, we discuss approximate treatments based on truncating the Hamiltonian in the spin-layer space and benchmark our results against a numerical solution of the Bethe-Salpeter equation finding good agreement. Our analysis uncovers a series of bright and dark excitons with their optical response depending on the combined spin-layer pseudospin conservation. We expect that the framework presented here will be useful in qualitatively understanding excitons in van der Waals magnets and guide the search for further excitons that still await discovery.

The article is organized as follows. Section II introduces the general s-d Hamiltonian, its reduction for systems with a magnetically ordered state, the CrSBr bilayer model, and the scope, limitations, and extensions of the model. Section III derives symmetry properties, the exact Γ -point eigenstates and energies, and the analytical direct-gap expressions. Section IV derives the optical selection rules and band-edge matrix elements. Section V formulates the layer-resolved Bethe-Salpeter equation and develops the analytical two-channel model for excitons, its layer decomposition, and the optical-response formalism. Section VI presents the numerical Bethe-Salpeter spectra, the field-dependent layer redistribution, and the dependence of the binding energy on screening, layer separation, and mass anisotropy. Section VII interprets the physical results and summarizes their experimental relevance and model scope.

II. GENERALIZED S-D MODEL EMPLOYING TIGHT-BINDING FRAMEWORK

We first introduce a general s-d Hamiltonian for itinerant electrons coupled to localized magnetic moments. We then obtain a single-particle Hamiltonian for a prescribed magnetically ordered state by replacing the localized-spin operators with their expectation values. We next specialize this Hamiltonian to the layer, orbital, and spin degrees of freedom relevant to the near-gap bands of a CrSBr bilayer. Finally, we outline the scope of the model, including more general multilayers and magnetic textures, and the extensions required to incorporate additional orbitals, interlayer tunneling, spin-orbit coupling, and dynamical-spin effects.

A. Generalized s-d Hamiltonian and ordered-state reduction

Let $o = 1, \dots, N_{\text{orb}}$ label the low-energy effective electronic orbitals retained in layer $\ell = 1, \dots, N_L$, and let $\hat{c}_{\mathbf{R}\ell os}$ annihilate an electron with spin s in the in-plane unit cell $\mathbf{R}$. The $2N_{\text{orb}}$ -component spinor $\hat{\mathbf{c}}_{\mathbf{R}\ell}$ collects the orbital and electron-spin indices. The Hamiltonian of the generalized s-d model is

$$\begin{aligned}\hat{H} &= \hat{H}_{\text{el}} + \hat{H}_{\text{ex}} + \hat{H}_{\text{spin}}, \\ \hat{H}_{\text{el}} &= \sum_{\mathbf{R}\mathbf{R}'} \sum_{\ell\ell'} \hat{\mathbf{c}}_{\mathbf{R}\ell}^\dagger \mathcal{T}_{\mathbf{R}\mathbf{R}'}^{\ell\ell'} \hat{\mathbf{c}}_{\mathbf{R}'\ell'}, \\ \hat{H}_{\text{ex}} &= - \sum_{\mathbf{R}\ell o} \sum_{\nu, \nu'=x,y,z} \hat{s}_{\mathbf{R}\ell o}^\nu \mathcal{J}_{\ell o}^{\nu\nu'} \hat{s}_{\mathbf{R}\ell}^{\nu'}, \\ \hat{H}_{\text{spin}} &= \frac{1}{2} \sum_{\mathbf{R}\mathbf{R}'} \sum_{\ell\ell'} \hat{\mathbf{S}}_{\mathbf{R}\ell}^T \mathcal{K}_{\mathbf{R}\mathbf{R}'}^{\ell\ell'} \hat{\mathbf{S}}_{\mathbf{R}'\ell'} + \mu_B \sum_{\mathbf{R}\ell} \mathbf{B}_{\text{ext}}^T \mathbf{g}_\ell \hat{\mathbf{S}}_{\mathbf{R}\ell}.\end{aligned}\tag{1}$$

The matrix $\mathcal{T}_{\mathbf{R}\mathbf{R}'}^{\ell\ell'}$ acts in the combined orbital and electron-spin space. It contains the on-site energies, intralayer hopping, interlayer tunneling, orbital mixing, and spin-dependent hopping. The term $\hat{H}_{\text{ex}}$ describes the generalized s-d exchange interaction between the electrons and localized spins. The electron-spin density in orbital o is

$$\hat{s}_{\mathbf{R}\ell o} = \frac{1}{2} \sum_{ss'} \hat{\mathbf{c}}_{\mathbf{R}\ell os}^\dagger \boldsymbol{\sigma}_{ss'} \hat{\mathbf{c}}_{\mathbf{R}\ell os'},\tag{2}$$

and $\hat{\mathbf{S}}_{\mathbf{R}\ell}$ is the dimensionless localized-spin operator, such that the corresponding angular-momentum operator is $\hbar \hat{\mathbf{S}}_{\mathbf{R}\ell}$. The exchange tensor $\mathcal{J}_{\ell o}^{\nu\nu'}$ may depend on the orbital and layer and contain both isotropic and anisotropic exchange couplings. The 3×3 tensor $\mathcal{K}_{\mathbf{R}\mathbf{R}'}^{\ell\ell'}$ describes the localized-spin interactions, including exchange and anisotropy. The second term in $\hat{H}_{\text{spin}}$ is the Zeeman coupling to the applied magnetic field $\mathbf{B}_{\text{ext}}$, where μ_B is the Bohr magneton and $\mathbf{g}_\ell$ is the localized-spin g tensor.

For a prescribed magnetically ordered state, we replace each localized-spin operator by its expectation value,

$$\hat{\mathbf{S}}_{\mathbf{R}\ell} \longrightarrow S_\ell \mathbf{n}_\ell(\mathbf{R}), \quad |\mathbf{n}_\ell(\mathbf{R})| = 1.\tag{3}$$

Here, S_ℓ is the magnitude of the ordered spin expectation value and $\mathbf{n}_\ell(\mathbf{R})$ specifies its direction in layer ℓ and in-plane unit cell $\mathbf{R}$. The exchange tensor then generates the local electronic exchange field

$$\mathbf{h}_{\ell o}(\mathbf{R}) = S_\ell \mathcal{J}_{\ell o} \mathbf{n}_\ell(\mathbf{R}),\tag{4}$$

where $(\mathcal{J}_{\ell o})_{\nu\nu'} = \mathcal{J}_{\ell o}^{\nu\nu'}$. The electron-spin interaction becomes

$$\hat{H}_{\text{ex}}^{\text{stat}} = -\frac{1}{2} \sum_{\mathbf{R}\ell o} \hat{\mathbf{c}}_{\mathbf{R}\ell o}^\dagger [\mathbf{h}_{\ell o}(\mathbf{R}) \cdot \boldsymbol{\sigma}] \hat{\mathbf{c}}_{\mathbf{R}\ell o}.\tag{5}$$

For isotropic exchange, $\mathcal{J}_{\ell o}^{\nu\nu'} = \mathcal{J}_\ell^\circ \delta_{\nu\nu'}$, we define the exchange-energy scale $J_\ell^\circ = \mathcal{J}_\ell^\circ S_\ell / 2$. Equation (5) then reduces to $-\sum_{\mathbf{R}\ell o} J_\ell^\circ \hat{\mathbf{c}}_{\mathbf{R}\ell o}^\dagger [\mathbf{n}_\ell(\mathbf{R}) \cdot \boldsymbol{\sigma}] \hat{\mathbf{c}}_{\mathbf{R}\ell o}$.

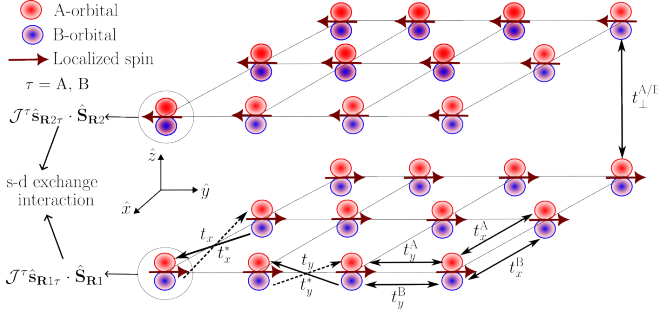


FIG. 1. Schematic of the reduced low-energy model for a magnetically ordered CrSBr bilayer. Each layer $\ell = 1, 2$ contains two effective orbital sectors, $\tau = A, B$, representing the conduction- and valence-band-edge manifolds, respectively. The localized Cr spin $\hat{\mathbf{S}}_{\mathbf{R}\ell}$ couples to the electron spin $\hat{\mathbf{S}}_{\mathbf{R}\ell\tau}$ through the orbital-dependent s-d exchange interaction $-\mathcal{J}^{\tau} \hat{\mathbf{S}}_{\mathbf{R}\ell\tau} \cdot \hat{\mathbf{S}}_{\mathbf{R}\ell}$. The parameters t_x^{τ} and t_y^{τ} describe intralayer hopping along the crystallographic a and b directions, respectively, while $t_{\perp}^{\tau}$ denotes spin-conserving interlayer hopping between equivalent orbital sectors. The interorbital hoppings are denoted by t_x and t_y , with $t_x = 0$ and $t_y = ig_y/2$. Red and blue circles represent the effective A- and B-orbital sectors, respectively, and the arrows indicate the localized Cr spins.

This construction applies to multilayer systems with collinear, canted, or noncollinear magnetic order. Crystal momentum is well defined when the electronic coefficients and the magnetic order share a common translation group. For an N_L -layer system with nearest-neighbor hopping along the stacking direction, the corresponding Bloch Hamiltonian takes the block form

$$H_{N_L}(\mathbf{k}) = \begin{pmatrix} h_1(\mathbf{k}) & \mathbf{T}_{12}(\mathbf{k}) & 0 & \cdots \\ \mathbf{T}_{12}^{\dagger}(\mathbf{k}) & h_2(\mathbf{k}) & \mathbf{T}_{23}(\mathbf{k}) & \ddots \\ 0 & \mathbf{T}_{23}^{\dagger}(\mathbf{k}) & \ddots & \ddots \\ \vdots & \ddots & \ddots & h_{N_L}(\mathbf{k}) \end{pmatrix}, \quad (6)$$

where each layer block has dimension $2N_{\text{orb}}$. Here, $h_{\ell}(\mathbf{k})$ contains the intralayer electronic terms and the exchange field in layer ℓ , while $\mathbf{T}_{\ell,\ell+1}(\mathbf{k})$ describes hopping between adjacent layers.

B. Reduction to a CrSBr bilayer

For CrSBr, we specialize the generic orbital space to two effective orbital sectors, A and B, that span the conduction- and valence-edge manifolds after the remote bands have been integrated out. We choose $x \parallel a$, $y \parallel b$, and $z \parallel c$. The b axis is the magnetic easy axis, whereas c is the hard out-of-plane axis. Figure 1 shows the degrees of freedom retained in the reduced bilayer model.

Within the reduced model, interlayer hopping preserves both the effective orbital sector and the electron spin. We retain nearest-neighbor intralayer hopping along the crystallographic a and b directions. For the interorbital sector, we retain the leading coupling along b , consistent

with the near- Γ description of the CrSBr band edges and their strongly b -polarized optical response [33–35].

Let $\mathbf{m}_1$ and $\mathbf{m}_2$ be unit vectors along the uniform ordered moments in the two layers, with $|\mathbf{m}_{\ell}| = 1$. The two layers are taken to be electronically equivalent, while their magnetic configurations are specified by $\mathbf{m}_{\ell}$. In the isotropic ordered-state limit of Eq. (5), the exchange-energy scale is orbital dependent, $J_{\ell}^o \rightarrow J^{\tau}$. We denote the orbital opposite to τ by $\bar{\tau}$, with $\bar{A} = B$ and $\bar{B} = A$. The ordered-state electronic Hamiltonian is $\hat{H}_{\text{CrSBr}}^{\text{el}} = \hat{H}_{\text{el}}^{\text{CrSBr}} + \hat{H}_{\text{ex}}^{\text{CrSBr}}$, where

$$\begin{aligned} \hat{H}_{\text{el}}^{\text{CrSBr}} = & \sum_{\mathbf{R}\ell\tau s} \epsilon^{\tau} \hat{c}_{\mathbf{R}\ell\tau s}^{\dagger} \hat{c}_{\mathbf{R}\ell\tau s} \\ & - \sum_{\mathbf{R}\ell\tau s} \sum_{\nu=x,y} \sum_{\delta=\pm 1} t_{\nu}^{\tau} \hat{c}_{\mathbf{R}\ell\tau s}^{\dagger} \hat{c}_{\mathbf{R}+\delta\mathbf{a}_{\nu},\ell\tau s} \\ & - \sum_{\mathbf{R}\tau s} t_{\perp}^{\tau} \left(\hat{c}_{\mathbf{R}1\tau s}^{\dagger} \hat{c}_{\mathbf{R}2\tau s} + \hat{c}_{\mathbf{R}2\tau s}^{\dagger} \hat{c}_{\mathbf{R}1\tau s} \right) \\ & + \frac{ig_y}{2} \sum_{\mathbf{R}\ell\tau s} \hat{c}_{\mathbf{R}\ell\tau s}^{\dagger} \left(\hat{c}_{\mathbf{R}+\mathbf{a}_y,\ell\bar{\tau}s} - \hat{c}_{\mathbf{R}-\mathbf{a}_y,\ell\bar{\tau}s} \right). \end{aligned} \quad (7)$$

while the ordered-state s-d exchange is

$$\hat{H}_{\text{ex}}^{\text{CrSBr}} = - \sum_{\mathbf{R}\ell\tau} \sum_{ss'} J^{\tau} \hat{c}_{\mathbf{R}\ell\tau s}^{\dagger} (\mathbf{m}_{\ell} \cdot \boldsymbol{\sigma})_{ss'} \hat{c}_{\mathbf{R}\ell\tau s'}. \quad (8)$$

Here, $\mathbf{a}_{\nu} = a_{\nu} \hat{\nu}$ denotes the nearest-neighbor displacement along $\nu = x, y$. The parameters ϵ^{τ} , t_{ν}^{τ} , and $t_{\perp}^{\tau}$ denote the onsite, intralayer, and interlayer matrix elements of orbital sector τ , respectively, and specify the corresponding entries of $\mathcal{T}_{\mathbf{R}\mathbf{R}'}^{\ell\ell'}$ in Eq. (1). The sum over $\delta = \pm 1$ runs over the two nearest neighbors at $\mathbf{R} \pm \mathbf{a}_{\nu}$. The g_y term instead describes interorbital hopping along the crystallographic b direction with opposite signs for displacements $+\mathbf{a}_y$ and $-\mathbf{a}_y$, consistent with the strongly b -polarized interband coupling of CrSBr [33, 34]. Equation (8) is the CrSBr form of the ordered-state s-d exchange in Eq. (5).

For the momentum-space representation, we denote Pauli matrices in layer, orbital, and electron-spin space by η_i , τ_i , and σ_i , respectively. The layer projectors are $\Pi_{1,2} = (\eta_0 \pm \eta_z)/2$, and the orbital projectors are $\tau_{A,B} = (\tau_0 \pm \tau_z)/2$. We define the orbital-dependent exchange and interlayer hopping matrices as

$$J_{\text{orb}} = J^A \tau_A + J^B \tau_B, \quad \mathcal{T}_{\perp} = t_{\perp}^A \tau_A + t_{\perp}^B \tau_B. \quad (9)$$

With $\hat{c}_{\mathbf{R}\ell\tau s} = N^{-1/2} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{R}} \hat{c}_{\mathbf{k}\ell\tau s}$, the corresponding single-particle Bloch Hamiltonian is

$$\begin{aligned} H_{\text{CrSBr}}^{\text{orb}}(\mathbf{k}; \mathbf{m}_1, \mathbf{m}_2) = & \eta_0 \otimes h_{\text{orb}}(\mathbf{k}) \otimes \sigma_0 \\ & - \sum_{\ell=1,2} \Pi_{\ell} \otimes J_{\text{orb}} \otimes (\mathbf{m}_{\ell} \cdot \boldsymbol{\sigma}) \\ & - \eta_x \otimes \mathcal{T}_{\perp} \otimes \sigma_0, \end{aligned} \quad (10)$$

with

$$\begin{aligned} h_{\text{orb}}(\mathbf{k}) = & \epsilon_A(\mathbf{k}) \tau_A + \epsilon_B(\mathbf{k}) \tau_B + d_y(\mathbf{k}) \tau_x, \\ \epsilon_{\tau}(\mathbf{k}) = & \bar{\epsilon}^{\tau} + 2t_x^{\tau} [1 - \cos(k_x a_x)] + 2t_y^{\tau} [1 - \cos(k_y a_y)], \\ d_y(\mathbf{k}) = & -g_y \sin(k_y a_y). \end{aligned} \quad (11)$$

The real-space onsite energy is related to the band-edge reference energy by $\bar{\epsilon}^\tau = \epsilon^\tau - 2t_x^\tau - 2t_y^\tau$. The opposite signs of the interorbital hopping for displacements $\pm \mathbf{a}_y$ give $(ig_y/2)(e^{ik_y a_y} - e^{-ik_y a_y}) = -g_y \sin(k_y a_y)$, which yields $d_y(\mathbf{k})$ in Eq. (11). Near Γ -point, this term is linear in k_y and is the lattice counterpart of the leading $k_y p_{cv}$ interband coupling in low-energy descriptions of CrSBr [31, 33, 36].

The anisotropy between t_x^τ and t_y^τ reproduces the quasi-one-dimensional band curvature [31, 33, 36]. Because $d_y(\Gamma) = 0$, the two orbital sectors decouple at the Γ point. Its k_y derivative is nevertheless finite. The Bloch velocity operator is $\hat{V}_\nu(\mathbf{k}) = \hbar^{-1} \partial_{k_\nu} H_{\text{CrSBr}}^{\text{orb}}(\mathbf{k})$. In the layer-orbital-spin basis, its components have a matrix representation. At Γ -point, $\hat{V}_x = 0$, whereas $\hat{V}_y = -(a_y g_y / \hbar) \eta_0 \otimes \tau_x \otimes \sigma_0$. Hence, the b -directed velocity operator connects the A and B orbital sectors at the band edge, and its interband matrix elements determine the b -polarized optical coupling. This is the origin of the strong linear dichroism of CrSBr [33, 34].

Equation (10) does not select a direction for the applied magnetic field. The field acts on the localized moments through $\hat{H}_{\text{spin}}$, and minimization of the magnetic free energy determines $\mathbf{m}_1$ and $\mathbf{m}_2$.

For the analytical results developed below, we consider a magnetic field along the crystallographic c axis. Minimization of the two-sublattice magnetic free energy gives the symmetric canting trajectory

$$\begin{aligned} \mathbf{m}_1 &= (0, \cos \alpha, \sin \alpha), \\ \mathbf{m}_2 &= (0, -\cos \alpha, \sin \alpha), \\ \sin \alpha &= \frac{B_{\text{ext}}}{B_{\text{sat}}}, \quad 0 \leq B_{\text{ext}} \leq B_{\text{sat}}. \end{aligned} \quad (12)$$

Here, α is the canting angle, B_{ext} is the applied field, and B_{sat} is the saturation field. Appendix A derives this trajectory from the magnetic free energy.

At zero field ($\alpha = 0$), the ordered layer moments are antiparallel (AP) along the easy b axis, corresponding to interlayer antiferromagnetic (AFM) order. For $0 < B_{\text{ext}} < B_{\text{sat}}$ ($0 < \alpha < \pi/2$), the equilibrium state is a canted AFM configuration with the b -axis components remaining antiparallel, whereas the c -axis components are parallel to the applied field. At $B_{\text{ext}} = B_{\text{sat}}$ ($\alpha = \pi/2$), the b -axis components vanish and the two moments become parallel (P) along c , producing the field-polarized interlayer ferromagnetic (FM) configuration. Above saturation, the moments remain aligned along c . Figure 2 summarizes this field-driven evolution of the magnetic order.

Substituting the equilibrium moment directions in Eq. (12) into Eq. (10) gives

$$\begin{aligned} H_{\text{CrSBr}}(\mathbf{k}) &= \eta_0 \otimes h_{\text{orb}}(\mathbf{k}) \otimes \sigma_0 \\ &\quad - \eta_0 \otimes J_{\text{orb}} \otimes \sin \alpha \sigma_z \\ &\quad - \eta_z \otimes J_{\text{orb}} \otimes \cos \alpha \sigma_y \\ &\quad - \eta_x \otimes \mathcal{T}_\perp \otimes \sigma_0. \end{aligned} \quad (13)$$

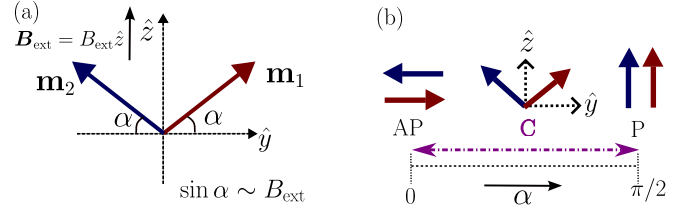


FIG. 2. Field-induced canting of the ordered moments in a CrSBr bilayer for an applied magnetic field $B_{\text{ext}} \parallel c$. (a) Definition of the symmetric canting configuration. The ordered moment directions $\mathbf{m}_1$ and $\mathbf{m}_2$ have antiparallel components along the magnetic easy axis b and equal components along the applied-field direction c . The canting angle α is measured from the b axis. (b) Evolution of the magnetic configuration from the AFM state at $\alpha = 0$, through the canted regime, to the field-polarized state at $\alpha = \pi/2$. AP and P denote antiparallel and parallel layer-moment configurations, respectively.

The first line is the spin-independent intralayer band Hamiltonian. The second line is the uniform c -axis component of the exchange field, which has the same sign in both layers. The third line is the staggered b -axis component, which has opposite signs in the two layers. The last line is the interlayer tunneling. At $\alpha = 0$, the uniform component vanishes and the exchange field is purely staggered. At $\alpha = \pi/2$, the staggered component vanishes and the exchange field is uniform.

The hopping matrix is field independent in the present model. The magnetic field enters the electronic Hamiltonian through the equilibrium canting angle and changes the layer and spin composition of the Bloch states, thereby modifying their effective interlayer hybridization. The minimal electronic Hamiltonian does not include a separate bare Zeeman term for the itinerant electrons. The orbital dependence of J^τ and $t_\perp^\tau$ accounts for the different exchange splittings and interlayer hybridization of the conduction and valence edges [4, 12, 18, 35]. Appendix B relates the reduced lattice parameters to the band-edge masses and optical matrix elements.

C. Generality, scope, and extensions

Equation (1) provides a general s-d framework for layered magnetic systems with arbitrary orbital content, layer number, hopping structure, and magnetic configuration. For compactness, we introduce one effective localized moment per in-plane unit cell and layer. Systems with multiple magnetic sublattices can be incorporated by adding a corresponding site index to $\hat{\mathbf{S}}$ and $\mathcal{J}$. Different materials and stacking geometries can be described by the appropriate choices of N_{orb} , N_L , $\mathcal{J}$, and $\mathcal{T}$.

Equation (10) is the low-energy realization of this framework for a CrSBr bilayer. It retains two effective near direct gap orbital sectors, equivalent layers, a uniform ordered moment in each layer, orbital-dependent isotropic electron-spin exchange, and orbital-preserving

spin-conserving interlayer hopping. Remote electronic bands are incorporated through the effective parameters of the reduced Hamiltonian. Equation (13) further specializes this model to the symmetric c -axis canting trajectory defined in Eq. (12).

The general Hamiltonian can also describe noncollinear and spatially varying magnetic textures. A Bloch representation applies when the electronic structure and magnetic order share a common translation group. Commensurate magnetic or moiré modulations can then be treated using an enlarged supercell or a continuum moiré basis, whereas incommensurate structures require a real-space formulation. Inequivalent layers, additional orbital sectors, twist-dependent tunneling, spin-dependent hopping, and stronger spin-orbit interactions can be included by enlarging the basis and extending the corresponding hopping and exchange matrices.

Replacing the localized-spin operators by their ordered expectation values treats the magnetic texture as static on the electronic time scale. Quantum and dynamical spin fluctuations are therefore not included explicitly in the present single-particle Hamiltonian, but they can be incorporated through a dynamical treatment of the localized moments and their coupling to the electrons.

III. SPIN-RESOLVED LAYER PARITY SYMMETRY AND BAND STRUCTURE

We now focus on the CrSBr bilayer described by Eq. (13). Magnetic canting changes the layer and spin composition of the band-edge states and thereby modifies their interlayer hybridization. Along the symmetric c -axis canting trajectory, however, the Hamiltonian retains a composite layer-spin symmetry throughout the evolution from the antiferromagnetic state to the field-polarized state. This symmetry provides a natural basis for diagonalizing Eq. (13) and labeling its eigenstates.

A. Conserved spin-resolved layer-parity symmetry

Along the symmetric canting trajectory in Eq. (12), no fixed Cartesian spin component is conserved for $0 < \alpha < \pi/2$. Nevertheless, the composite operator

$$\zeta = \eta_x \otimes \tau_0 \otimes \sigma_z \quad (14)$$

commutes with the Bloch Hamiltonian, $[\zeta, H_{\text{CrSBr}}(\mathbf{k})] = 0$ for $0 \leq \alpha \leq \pi/2$. The symmetry follows from the combined action of the layer and spin factors in ζ [see the proof in Appendix C]. The layer-exchange operator η_x changes the sign of η_z , while σ_z changes the sign of σ_y . These two sign changes cancel in the staggered exchange term $\eta_z \otimes J_{\text{orb}} \otimes \sigma_y$. The operator ζ also commutes separately with the orbital, uniform-exchange, and interlayer-hopping terms in Eq. (13).

We refer to ζ as the spin-resolved layer-parity (SRLP) operator. Since $\zeta^2 = \mathbb{I}_8$, its eigenvalues are $\lambda = \pm 1$, and

the eight-dimensional single-particle Hilbert space separates into two four-dimensional sectors $\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_-$. Each Bloch eigenstate $|\xi_\lambda(\mathbf{k})\rangle$ can be chosen as a simultaneous eigenstate of $H_{\text{CrSBr}}(\mathbf{k})$ and ζ , with $\zeta |\xi_\lambda(\mathbf{k})\rangle = \lambda |\xi_\lambda(\mathbf{k})\rangle$. The SRLP eigenvalue remains a good quantum number in the canted regime, even though spin and layer parity are not separately conserved.

To make the composite character of λ explicit, we define the layer-parity states

$$|p_\pm\rangle = \frac{|1\rangle \pm |2\rangle}{\sqrt{2}}, \quad \eta_x |p_\pm\rangle = \pm |p_\pm\rangle. \quad (15)$$

For each orbital basis state $|\tau\rangle$, a convenient basis for the two SRLP sectors is

$$\begin{aligned} \mathcal{H}_+ : & \quad |p_+, \tau, \uparrow_z\rangle, \quad |p_-, \tau, \downarrow_z\rangle, \\ \mathcal{H}_- : & \quad |p_-, \tau, \uparrow_z\rangle, \quad |p_+, \tau, \downarrow_z\rangle, \quad \tau = \text{A, B}. \end{aligned} \quad (16)$$

Thus, the SRLP eigenvalue is the product of the layer-parity eigenvalue and the σ_z eigenvalue of each basis state.

The arrows in Eq. (16) label eigenstates of σ_z . For a general canting angle, they are basis labels rather than separately conserved quantum numbers. At the field-polarized state, σ_z and η_x are separately conserved. At the antiferromagnetic state, the conserved spin component is instead σ_y .

A perturbation preserves the SRLP decomposition only if it commutes with ζ . Spin-dependent hopping, spin-flip tunneling, and lower-symmetry spin-orbit terms can in general mix the two SRLP sectors. In that case, λ is no longer an exact quantum number.

B. Band-labeling convention

Within each SRLP sector, we label the four bands according to their energies at the Γ point:

$$E_{v'_\lambda}(\Gamma) < E_{v_\lambda}(\Gamma) < E_{c_\lambda}(\Gamma) < E_{c'_\lambda}(\Gamma). \quad (17)$$

Here, v_λ and v'_λ denote the upper and lower valence bands, respectively, while c_λ and c'_λ denote the lower and upper conduction bands. The SRLP eigenvalue $\lambda = \pm 1$ is the symmetry label of the state. The symbols c , c' , v , and v' specify the energy ordering within each SRLP sector and do not denote a fixed spin character.

This convention allows the bands to be followed continuously as the magnetic configuration changes. It also avoids ambiguities when the exchange couplings and interlayer hopping amplitudes differ between the conduction-like A sector and the valence-like B sector.

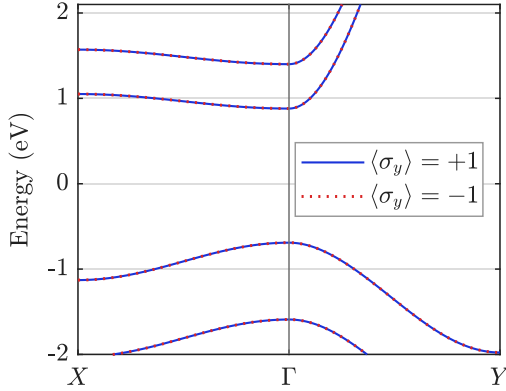
C. Antiferromagnetic and ferromagnetic limits

At the two collinear states, the exchange interaction acts as a Zeeman-like field along a fixed spin axis. In

(a) Schematics of state evolution for AFM bilayer at Γ -point

$$\begin{aligned}
 |\eta\rangle \otimes |\tau\rangle \otimes |\sigma\rangle &\equiv |\eta, \tau, \sigma\rangle \quad \eta \in \{1, 2\} \quad \tau \in \{A, B\} \quad \sigma \in \{\uparrow, \downarrow\} \quad |\uparrow_y\rangle = \frac{|\uparrow\rangle + i|\downarrow\rangle}{\sqrt{2}} \quad |\downarrow_y\rangle = \frac{|\uparrow\rangle - i|\downarrow\rangle}{\sqrt{2}} \\
 \begin{array}{c} \text{---} \text{---} \text{---} \text{---} \epsilon^A \\ |1, A, \uparrow\rangle \\ |2, A, \uparrow\rangle \end{array} & \quad \begin{array}{c} \text{---} \text{---} \epsilon^A + J^A \\ |1, A, \downarrow_y\rangle \\ |2, A, \uparrow_y\rangle \end{array} & \quad \begin{array}{c} \text{---} \text{---} \epsilon^A + \sqrt{(J^A)^2 + (t_\perp^A)^2} \\ |-1, A, \uparrow_y\rangle = \beta'_{1A\uparrow_y}|1, A, \uparrow_y\rangle - \beta'_{2A\uparrow_y}|2, A, \uparrow_y\rangle \\ |-1, A, \downarrow_y\rangle = \beta'_{1A\downarrow_y}|1, A, \downarrow_y\rangle - \beta'_{2A\downarrow_y}|2, A, \downarrow_y\rangle \end{array} \\
 \begin{array}{c} \text{---} \text{---} \epsilon^A \\ |1, A, \uparrow\rangle \\ |2, A, \uparrow\rangle \end{array} & \quad \begin{array}{c} \text{---} \text{---} \epsilon^A - J^A \\ |1, A, \uparrow_y\rangle \\ |2, A, \downarrow_y\rangle \end{array} & \quad \begin{array}{c} \text{---} \text{---} \epsilon^A - \sqrt{(J^A)^2 + (t_\perp^A)^2} \\ |+1, A, \uparrow_y\rangle = \beta_{1A\uparrow_y}|1, A, \uparrow_y\rangle + \beta_{2A\uparrow_y}|2, A, \uparrow_y\rangle \\ |+1, A, \downarrow_y\rangle = \beta_{1A\downarrow_y}|1, A, \downarrow_y\rangle + \beta_{2A\downarrow_y}|2, A, \downarrow_y\rangle \end{array} \\
 \dots\dots\dots & & \\
 \begin{array}{c} \text{---} \text{---} \text{---} \text{---} \epsilon^B \\ |1, B, \uparrow\rangle \\ |2, B, \uparrow\rangle \end{array} & \quad \begin{array}{c} \text{---} \text{---} \epsilon^B + J^B \\ |1, B, \uparrow_y\rangle \\ |2, B, \downarrow_y\rangle \end{array} & \quad \begin{array}{c} \text{---} \text{---} \epsilon^B + \sqrt{(J^B)^2 + (t_\perp^B)^2} \\ |-1, B, \uparrow_y\rangle = \beta_{1B\uparrow_y}|1, B, \uparrow_y\rangle - \beta_{2B\uparrow_y}|2, B, \uparrow_y\rangle \\ |-1, B, \downarrow_y\rangle = \beta_{1B\downarrow_y}|1, B, \downarrow_y\rangle - \beta_{2B\downarrow_y}|2, B, \downarrow_y\rangle \end{array} \\
 \begin{array}{c} \text{---} \text{---} \epsilon^B \\ |1, B, \uparrow\rangle \\ |2, B, \uparrow\rangle \end{array} & \quad \begin{array}{c} \text{---} \text{---} \epsilon^B - J^B \\ |1, B, \downarrow_y\rangle \\ |2, B, \uparrow_y\rangle \end{array} & \quad \begin{array}{c} \text{---} \text{---} \epsilon^B - \sqrt{(J^B)^2 + (t_\perp^B)^2} \\ |+1, B, \uparrow_y\rangle = \beta'_{1B\uparrow_y}|1, B, \uparrow_y\rangle + \beta'_{2B\uparrow_y}|2, B, \uparrow_y\rangle \\ |+1, B, \downarrow_y\rangle = \beta'_{1B\downarrow_y}|1, B, \downarrow_y\rangle + \beta'_{2B\downarrow_y}|2, B, \downarrow_y\rangle \end{array} \\
 J^{A/B} = t_\perp^{A/B} = 0 & \quad J^{A/B} \neq 0, t_\perp^{A/B} = 0 & \quad J^{A/B} \neq 0, t_\perp^{A/B} \neq 0
 \end{aligned}$$

(b) Band Structure for AFM Bilayer ($t_\perp^{A/B} = 0$)



(c) Band Structure for AFM Bilayer ($t_\perp^{A/B} \neq 0$)

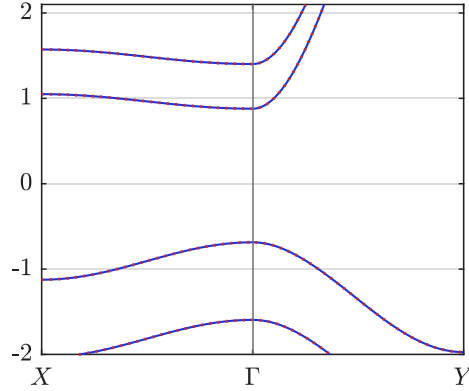


FIG. 3. Single-particle states and band structure of the interlayer antiferromagnetic bilayer at $\alpha = 0$. (a) Evolution of the Γ -point states as the orbital-dependent exchange and interlayer hopping are introduced. The left column shows the layer-degenerate states for $J^\tau = t_\perp^\tau = 0$; the middle column shows the layer-staggered exchange splitting for $J^\tau \neq 0$ and $t_\perp^\tau = 0$; and the right column shows the hybridized states for finite J^τ and $t_\perp^\tau$, with $\tau = A, B$. In this case, the exchange term $-\eta_z \otimes J_{\text{orb}} \otimes \sigma_y$ produces opposite Zeeman-like shifts in the two layers. For a fixed σ_y eigenvalue, interlayer hopping therefore couples layer states separated in energy by $2|J^\tau|$. The wave functions mix to first order in $t_\perp^\tau/J^\tau$, whereas the leading energy shift is second order, with scale $(t_\perp^\tau)^2/|J^\tau|$. (b) Band structure along $X-\Gamma-Y$ for $t_\perp^\tau = 0$. (c) Corresponding band structure for finite $t_\perp^\tau$. Solid and dotted curves denote states with $\langle\sigma_y\rangle = +1$ and -1 , respectively. The labels 1, 2, A, B, $\uparrow, \downarrow$, and $\uparrow_y, \downarrow_y$ denote layer states, orbital sectors, σ_z basis states, and σ_y eigenstates, respectively. Energies are shown in eV, and the parameters are listed in Appendix D.

the AFM state, this exchange field has opposite signs in the two layers. In the field-polarized FM state, it has the same sign in both layers. This difference determines whether interlayer hopping produces a second-order or first-order change in the band-edge energies.

1. Antiferromagnetic limit

At $\alpha = 0$, the ordered layer moments are antiparallel along the b axis. The exchange term in Eq. (13) becomes $-\eta_z \otimes J_{\text{orb}} \otimes \sigma_y$. It therefore acts as a layer-staggered Zeeman-like exchange field along y , with its orbital-dependent sign and magnitude contained in J^τ . The Hamiltonian commutes with σ_y . At the Γ point,

(a) Schematics of state evolution for FM bilayer at Γ -point

$$|\eta\rangle \otimes |\tau\rangle \otimes |\sigma\rangle \equiv |\eta, \tau, \sigma\rangle$$

$$\eta \in \{1, 2\} \tau \in \{A, B\} \sigma \in \{\uparrow, \downarrow\}$$

$$\begin{array}{cc} \text{---} & \text{---} \\ |1, A, \uparrow\rangle & |1, A, \downarrow\rangle \\ |2, A, \uparrow\rangle & |2, A, \downarrow\rangle \end{array} \epsilon^A$$

$$\begin{array}{cc} \text{---} & \text{---} \\ |1, A, \downarrow\rangle & \\ |2, A, \downarrow\rangle & \end{array} \epsilon^A + J^A$$

$$\begin{array}{cc} \text{---} & \text{---} \\ |1, A, \uparrow\rangle & \\ |2, A, \uparrow\rangle & \end{array} \epsilon^A - J^A$$

$$\text{---} \epsilon^A + J^A + t_\perp^A$$

$$|-1, A, \downarrow\rangle = \frac{|1\rangle - |2\rangle}{\sqrt{2}} \otimes |A\rangle \otimes |\downarrow\rangle$$

$$\text{---} \epsilon^A + J^A - t_\perp^A$$

$$|+1, A, \downarrow\rangle = \frac{|1\rangle + |2\rangle}{\sqrt{2}} \otimes |A\rangle \otimes |\downarrow\rangle$$

$$\text{---} \epsilon^A - J^A + t_\perp^A$$

$$|-1, A, \uparrow\rangle = \frac{|1\rangle - |2\rangle}{\sqrt{2}} \otimes |A\rangle \otimes |\uparrow\rangle$$

$$\text{---} \epsilon^A - J^A - t_\perp^A$$

$$|+1, A, \uparrow\rangle = \frac{|1\rangle + |2\rangle}{\sqrt{2}} \otimes |A\rangle \otimes |\uparrow\rangle$$

$$\begin{array}{cc} \text{---} & \text{---} \\ |1, B, \uparrow\rangle & |1, B, \downarrow\rangle \\ |2, B, \uparrow\rangle & |2, B, \downarrow\rangle \end{array} \epsilon^B$$

$$\begin{array}{cc} \text{---} & \text{---} \\ |1, B, \uparrow\rangle & \\ |2, B, \uparrow\rangle & \end{array} \epsilon^B + J^B$$

$$\begin{array}{cc} \text{---} & \text{---} \\ |1, B, \downarrow\rangle & \\ |2, B, \downarrow\rangle & \end{array} \epsilon^B - J^B$$

$$\text{---} \epsilon^B + J^B + t_\perp^B$$

$$|-1, B, \uparrow\rangle = \frac{|1\rangle - |2\rangle}{\sqrt{2}} \otimes |B\rangle \otimes |\uparrow\rangle$$

$$\text{---} \epsilon^B + J^B - t_\perp^B$$

$$|+1, B, \uparrow\rangle = \frac{|1\rangle + |2\rangle}{\sqrt{2}} \otimes |B\rangle \otimes |\uparrow\rangle$$

$$\text{---} \epsilon^B - J^B + t_\perp^B$$

$$|-1, B, \downarrow\rangle = \frac{|1\rangle - |2\rangle}{\sqrt{2}} \otimes |B\rangle \otimes |\downarrow\rangle$$

$$\text{---} \epsilon^B - J^B - t_\perp^B$$

$$|+1, B, \downarrow\rangle = \frac{|1\rangle + |2\rangle}{\sqrt{2}} \otimes |B\rangle \otimes |\downarrow\rangle$$

$$J^{A/B} = t_\perp^{A/B} = 0$$

$$J^{A/B} \neq 0, t_\perp^{A/B} = 0$$

$$J^{A/B} \neq 0, t_\perp^{A/B} \neq 0$$

(b) Band Structure for FM Bilayer ($t_\perp^{A/B} = 0$)

(c) Band Structure for FM Bilayer ($t_\perp^{A/B} \neq 0$)

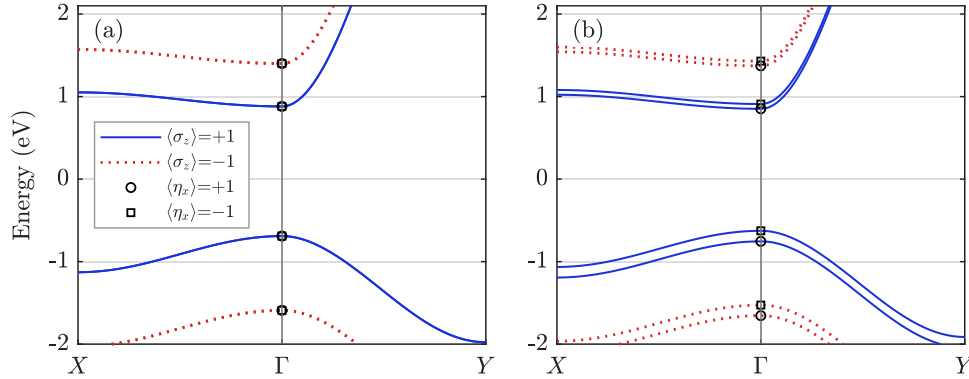


FIG. 4. Single-particle states and band structure of the field-polarized interlayer-ferromagnetic bilayer at $\alpha = \pi/2$. (a) Evolution of the Γ -point states as the orbital-dependent exchange and interlayer hopping are introduced. The left column shows the layer-degenerate states for $J^\tau = t_\perp^\tau = 0$; the middle column shows the exchange-split states for $J^\tau \neq 0$ and $t_\perp^\tau = 0$; and the right column shows the bonding and antibonding states for finite J^τ and $t_\perp^\tau$, with $\tau = A, B$. In this case, the exchange term $-\eta_0 \otimes J_{\text{orb}} \otimes \sigma_z$ produces the same Zeeman-like shift in both layers. Consequently, for a fixed σ_z eigenvalue, the two layer states are degenerate before interlayer hopping is introduced. Spin-conserving hopping couples these states resonantly and produces eigenstates with layer parity $p = \pm 1$, first-order energy shifts $-p t_\perp^\tau$, and a splitting $2|t_\perp^\tau|$. (b) Band structure along $X-\Gamma-Y$ for $t_\perp^\tau = 0$. (c) Corresponding band structure for finite $t_\perp^\tau$. Solid and dotted curves denote states with $\sigma_z = +1$ and -1 , respectively. The labels 1, 2, A, B, $\uparrow, \downarrow$, and ± 1 denote eigenstates or eigenvalues of η_z , τ_z , σ_z , and η_x , respectively. Energies are shown in eV, and the parameters are listed in Appendix D.

$d_y(\Gamma) = 0$, so the A and B orbital sectors also decouple. The exact band-edge energies are

$$\begin{aligned} E_{c'_\lambda}^{\text{AFM}} &= \bar{\epsilon}^A + \sqrt{(J^A)^2 + (t_\perp^A)^2}, \\ E_{c_\lambda}^{\text{AFM}} &= \bar{\epsilon}^A - \sqrt{(J^A)^2 + (t_\perp^A)^2}, \end{aligned}$$

$$E_{v_\lambda}^{\text{AFM}} = \bar{\epsilon}^B + \sqrt{(J^B)^2 + (t_\perp^B)^2},$$

$$E_{v'_\lambda}^{\text{AFM}} = \bar{\epsilon}^B - \sqrt{(J^B)^2 + (t_\perp^B)^2}. \quad (18)$$

These energies are independent of λ . The resulting twofold degeneracy is exact within the minimal AFM Hamiltonian.

The unitary operator

$$\mathcal{U}_{\text{AFM}} = \eta_x \otimes \tau_0 \otimes \sigma_x \quad (19)$$

enforces this degeneracy because

$$[\mathcal{U}_{\text{AFM}}, H_{\text{CrSBr}}(\mathbf{k}, \alpha = 0)] = 0, \quad \{\mathcal{U}_{\text{AFM}}, \zeta\} = 0. \quad (20)$$

If $|\psi_\lambda\rangle$ is an eigenstate with energy E and SRLP eigenvalue λ , then $\mathcal{U}_{\text{AFM}}|\psi_\lambda\rangle$ is an orthogonal eigenstate with the same energy and SRLP eigenvalue $-\lambda$.

For a fixed σ_y eigenvalue, the opposite Zeeman-like exchange fields shift the corresponding states in the two layers in opposite directions. Before interlayer hopping is introduced, these same-spin states are separated in energy by $2|J^\tau|$. Interlayer hopping preserves spin and therefore couples two nondegenerate layer states. Its diagonal matrix element vanishes, so there is no first-order energy correction.

For $|t_\perp^\tau| \ll |J^\tau|$, the energies in orbital sector τ have the expansion

$$E_{\tau,\beta}^{\text{AFM}} = \bar{\epsilon}^\tau + \beta \left[|J^\tau| + \frac{(t_\perp^\tau)^2}{2|J^\tau|} \right] + O \left[\frac{(t_\perp^\tau)^4}{|J^\tau|^3} \right], \quad \beta = \pm 1. \quad (21)$$

The leading energy shift is therefore second order in the interlayer hopping, with scale $(t_\perp^\tau)^2/|J^\tau|$. The wave functions acquire an opposite-layer component already at first order in $t_\perp^\tau/J^\tau$, even though the energy changes only at second order. Figure 3(a) illustrates this off-resonant hybridization, while Figs. 3(b) and 3(c) compare the band structures without and with interlayer hopping.

2. Ferromagnetic limit

At $\alpha = \pi/2$, the ordered layer moments are parallel along the c axis. The exchange term becomes $-\eta_0 \otimes J_{\text{orb}} \otimes \sigma_z$. It therefore acts as a layer-uniform Zeeman-like exchange field along z , again with the orbital-dependent sign and magnitude contained in J^τ . The operators σ_z and η_x are separately conserved, and the SRLP eigenvalue is their product,

$$\lambda = sp, \quad (22)$$

where $s = \pm 1$ and $p = \pm 1$ are the eigenvalues of σ_z and η_x , respectively.

With the interlayer hopping convention of Eq. (13), the exact Γ -point energies are

$$\begin{aligned} E_{\tau,s,p}^{\text{FM}} &= \bar{\epsilon}^\tau - sJ^\tau - pt_\perp^\tau, \\ &= \bar{\epsilon}^\tau - sJ^\tau - \lambda s t_\perp^\tau. \end{aligned} \quad (23)$$

The spin eigenvalue s , layer-parity eigenvalue p , and SRLP eigenvalue $\lambda = sp$ are kept distinct because their relation to the ordered band labels depends on the signs and relative magnitudes of J^τ and $t_\perp^\tau$. For the CrSBr parameterization used here, the conduction-like and valence-like

sectors have exchange couplings of opposite sign, $J^A > 0$ and $J^B < 0$, consistent with the opposite exchange response of the near-gap conduction and valence states [4, 12, 18, 35]. Consequently, for a fixed σ_z eigenvalue, the conduction- and valence-band edges are shifted in opposite energy directions. Within a given orbital sector, however, the uniform exchange field produces the same Zeeman-like shift in both layers. The two layer states are therefore degenerate before interlayer hopping is introduced. Interlayer hopping must then be diagonalized within this degenerate layer subspace, producing bonding and antibonding states with first-order energy shifts $-pt_\perp^\tau$ and a splitting $2|t_\perp^\tau|$.

Figure 4(a) illustrates this resonant hybridization. Figures 4(b) and 4(c) show that switching on interlayer hopping splits each pair of same-spin layer states already in first order.

The two limits show how magnetic order controls interlayer hybridization. In the AFM state, the layer-staggered Zeeman-like exchange field detunes same-spin states in opposite layers, so interlayer hopping produces a second-order energy shift. In the field-polarized FM state, the layer-uniform exchange field leaves the same-spin layer states degenerate, so the same hopping produces a first-order bonding and antibonding splitting. The comparison between Figs. 3 and 4 directly demonstrates this change from off-resonant to resonant interlayer hybridization.

D. Exact Γ -point energies and direct gaps at arbitrary canting

We now consider the continuous evolution between the AFM and field-polarized FM limits. At an arbitrary canting angle, the exchange field contains a layer-uniform component along z and a layer-staggered component along y . At the Γ point, the odd interorbital coupling vanishes, $d_y(\Gamma) = 0$, so the A and B orbital sectors decouple. Resolving each orbital sector by SRLP reduces the eight-dimensional Hamiltonian to four independent 2×2 blocks. Appendix E gives these blocks and their normalized eigenstates.

For orbital sector $\tau = A, B$ and SRLP eigenvalue $\lambda = \pm 1$, define

$$\begin{aligned} R_{\tau\lambda}(\alpha) &= \sqrt{[t_\perp^\tau + \lambda J^\tau \sin \alpha]^2 + [J^\tau \cos \alpha]^2} \\ &= \sqrt{(J^\tau)^2 + (t_\perp^\tau)^2 + 2\lambda J^\tau t_\perp^\tau \sin \alpha}. \end{aligned} \quad (24)$$

The first term under the square root combines interlayer hopping with the layer-uniform z -directed exchange field within a fixed SRLP sector. The second term is the contribution of the layer-staggered y -directed exchange field. The quantity $2R_{\tau\lambda}$ is the exact splitting between the two states in the corresponding orbital and SRLP block.

The four Γ -point band energies in each SRLP sector

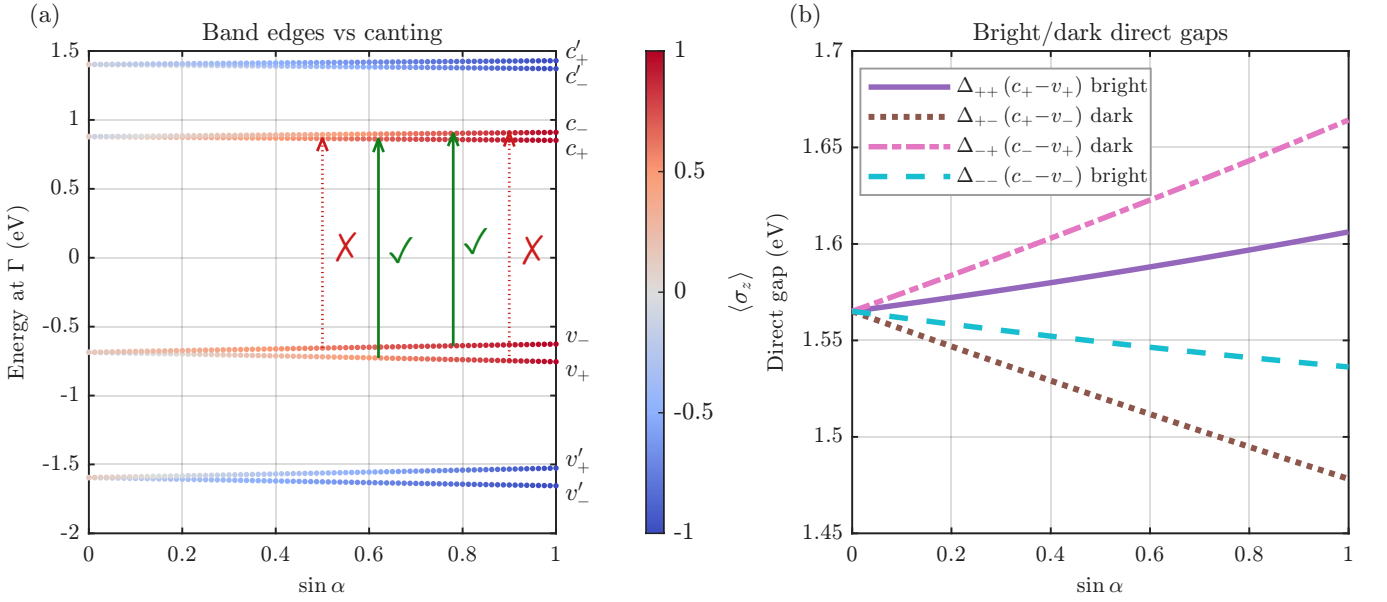


FIG. 5. Evolution of the Γ -point band edges and direct gaps along the symmetric c -axis canting trajectory. The horizontal coordinate is $\mathcal{B} = \sin \alpha = B_{\text{ext}}/B_{\text{sat}}$ below saturation. (a) Band-edge energies calculated from Eq. (25). Each band is labeled by its spin-resolved layer-parity (SRLP) eigenvalue $\lambda = \pm 1$, and the color scale represents $\langle \sigma_z \rangle$. At $\mathcal{B} = 0$, the exchange field is layer staggered and directed along y , and the bands form exactly degenerate SRLP partner pairs within the reduced Hamiltonian. At $\mathcal{B} = 1$, the exchange field is layer uniform and directed along z , and the states have definite σ_z . At intermediate canting, spin and layer parity are separately mixed, while SRLP remains conserved. Green solid arrows connect conduction and valence states with equal SRLP, whereas red dotted arrows connect states with opposite SRLP. Check marks and crosses denote symmetry-allowed and symmetry-forbidden transitions, respectively, according to the selection rule derived in Sec. IV. (b) The four direct gaps $\Delta_{\lambda\lambda'} = E_{c_\lambda}(\Gamma) - E_{v_{\lambda'}}(\Gamma)$, calculated from Eq. (26). Equal-SRLP gaps Δ_{++} and Δ_{--} correspond to the two lowest symmetry-allowed channels, whereas Δ_{+-} and Δ_{-+} correspond to opposite-SRLP channels. The numerical parameters are listed in Appendix D.

are

$$\begin{aligned} E_{c'_\lambda}(\Gamma) &= \bar{\epsilon}^A + R_{A\lambda}(\alpha), & E_{c_\lambda}(\Gamma) &= \bar{\epsilon}^A - R_{A\lambda}(\alpha), \\ E_{v_\lambda}(\Gamma) &= \bar{\epsilon}^B + R_{B\lambda}(\alpha), & E_{v'_{\lambda}}(\Gamma) &= \bar{\epsilon}^B - R_{B\lambda}(\alpha). \end{aligned} \quad (25)$$

These exact energies interpolate continuously between the two collinear limits. At $\alpha = 0$, $R_{\tau\lambda} = \sqrt{(J^\tau)^2 + (t^\tau_\perp)^2}$ is independent of λ , reproducing the exact AFM energies in Eq. (18). At $\alpha = \pi/2$, $R_{\tau\lambda} = |J^\tau + \lambda t^\tau_\perp|$, which reproduces the FM energies in Eq. (23) after applying the band-ordering convention in Eq. (17).

The four direct transitions formed from $v_{\lambda'}$ and c_λ have the single-particle gaps

$$\begin{aligned} \Delta_{\lambda\lambda'}(\alpha) &\equiv E_{c_\lambda}(\Gamma) - E_{v_{\lambda'}}(\Gamma) \\ &= \Delta_\Gamma^0 - R_{A\lambda}(\alpha) - R_{B\lambda'}(\alpha), \end{aligned} \quad (26)$$

where

$$\Delta_\Gamma^0 = \bar{\epsilon}^A - \bar{\epsilon}^B \quad (27)$$

is the separation between the bare A- and B-orbital energies at Γ . The pair of SRLP indices (λ, λ') labels four distinct direct transitions. At the AFM state, $R_{\tau,+} = R_{\tau,-}$, so all four gaps are degenerate within the minimal model. Canting lifts this degeneracy while each single-particle

state retains a definite SRLP eigenvalue. The resulting evolution of the band-edge energies and direct gaps is shown in Fig. 5.

To identify the leading canting dependence of the direct gaps, we write $\mathcal{B} = \sin \alpha$ and define

$$R_\tau = \sqrt{(J^\tau)^2 + (t^\tau_\perp)^2}, \quad r_\tau = \frac{2J^\tau t^\tau_\perp}{R_\tau^2}, \quad \tau = A, B. \quad (28)$$

Expanding the exact result in the dimensionless mixing parameters r_τ , with $\mathcal{B} = \sin \alpha$, gives

$$\Delta_{\lambda\lambda'}(\mathcal{B}) = \Delta'_\Gamma + \chi_{\lambda\lambda'} \mathcal{B} + \xi_{\text{sp}} \mathcal{B}^2 + O\left(\max_\tau R_\tau |r_\tau \mathcal{B}|^3\right), \quad (29)$$

where

$$\begin{aligned} \Delta'_\Gamma &= \Delta_\Gamma^0 - R_A - R_B, \\ \chi_{\lambda\lambda'} &= -\lambda \frac{J^A t^A_\perp}{R_A} - \lambda' \frac{J^B t^B_\perp}{R_B}, \\ \xi_{\text{sp}} &= \frac{(J^A t^A_\perp)^2}{2R_A^3} + \frac{(J^B t^B_\perp)^2}{2R_B^3}. \end{aligned} \quad (30)$$

The coefficient $\chi_{\lambda\lambda'}$ gives the channel-dependent linear variation with $\sin \alpha$, whereas the positive coefficient ξ_{sp} gives the leading quadratic single-particle contribution common to all four channels.

Equation (29) provides a controlled approximation to the exact gaps throughout the canting interval when $\max_{\tau=A,B} |r_\tau| \ll 1$, as occurs in the exchange-dominated regime $|t_\perp^\tau| \ll |J^\tau|$. The exact energies are given by Eq. (25), while Eq. (29) separates the leading linear and quadratic canting dependences. Its derivation is given in Appendix E.

Equations (25) through (30) provide the exact band-edge energies and direct gaps along the symmetric canting trajectory. They separate the channel-dependent linear detuning from the common quadratic single-particle contribution and allow each SRLP-resolved transition to be followed continuously between the AFM and field-polarized FM limits.

IV. SINGLE-PARTICLE OPTICAL EXCITATION SELECTION RULES

The band-edge optical response is determined by the orbital structure of the velocity operator and by the SRLP symmetry of the Bloch states. For valence and conduction eigenstates of Eq. (13), we define

$$M_{cv}^\nu(\mathbf{k}) = \langle c(\mathbf{k}) | \partial_{k_\nu} H_{\text{CrSBr}}(\mathbf{k}) | v(\mathbf{k}) \rangle = \hbar V_{cv}^\nu(\mathbf{k}). \quad (31)$$

Here, M_{cv}^ν is the interband matrix element of the momentum derivative of the Bloch Hamiltonian [37–39] and its magnitude largely governs the oscillator strength of the corresponding optical transition as discussed further below.

At the Γ point, the derivatives of the diagonal orbital dispersions vanish. The momentum-odd interorbital term $d_y(\mathbf{k}) = -g_y \sin(k_y a_y)$ therefore gives

$$\hat{V}_x(\Gamma) = 0, \quad \hat{V}_y(\Gamma) = -\frac{a_y g_y}{\hbar} \eta_0 \otimes \tau_x \otimes \sigma_0. \quad (32)$$

Thus, within the reduced model, the band-edge optical matrix element is finite for polarization along $y \parallel b$, whereas it vanishes for polarization along $x \parallel a$. The operator τ_x connects the A and B orbital sectors, while acting as the identity in layer and spin space. This orbital structure produces the strong linear dichroism characteristic of CrSBr [12, 18, 33, 34].

The SRLP symmetry supplies an additional exact selection rule. Because ζ is momentum independent and commutes with $H_{\text{CrSBr}}(\mathbf{k})$ throughout the symmetric canting trajectory,

$$[\zeta, \partial_{k_\nu} H_{\text{CrSBr}}(\mathbf{k})] = 0. \quad (33)$$

Consequently,

$$[\zeta, \hat{V}_\nu(\mathbf{k})] = 0, \quad \lambda_c \neq \lambda_v \implies \langle c_{\lambda_c}(\mathbf{k}) | \hat{V}_\nu(\mathbf{k}) | v_{\lambda_v}(\mathbf{k}) \rangle = 0. \quad (34)$$

An optical transition is therefore symmetry allowed only when the conduction and valence states have the same SRLP eigenvalue,

$$\lambda_c = \lambda_v. \quad (35)$$

In particular, the two lowest direct transitions $v_- \rightarrow c_-$ and $v_+ \rightarrow c_+$ are allowed, whereas $v_- \rightarrow c_+$ and $v_+ \rightarrow c_-$ are forbidden in the minimal Hamiltonian. Appendix F gives the formal proof and the exact band-edge matrix elements.

Equal SRLP is necessary but not sufficient for a finite optical matrix element. Because $\hat{V}_y(\Gamma)$ acts as the identity in the layer-spin subspace, the magnitude of an allowed matrix element is determined by the overlap between the layer-spin components of the corresponding A- and B-sector eigenstates. Destructive interference or orthogonality between these components can suppress an otherwise symmetry-allowed transition.

For the CrSBr parameters used here, the lowest equal-SRLP transitions carry substantial oscillator strength throughout the canting trajectory. The equal-SRLP cross transitions involving a c' or v' band are weaker but generally finite in the AFM state and at intermediate canting. At the field-polarized FM state, these cross transitions vanish because spin and layer parity become separately conserved. This state suppression is therefore stronger than the SRLP selection rule alone.

The two lowest symmetry-allowed direct gaps are

$$\begin{aligned} \Delta_\lambda^{\text{opt}}(\alpha) &= E_{c_\lambda}(\Gamma) - E_{v_\lambda}(\Gamma) \\ &= \Delta_\Gamma^0 - R_{A\lambda}(\alpha) - R_{B\lambda}(\alpha) \\ &\simeq \Delta'_\Gamma + \chi_{\lambda\lambda} \mathcal{B} + \xi_{\text{sp}} \mathcal{B}^2, \quad \mathcal{B} = \sin \alpha. \end{aligned} \quad (36)$$

The final line uses the expansion in Eq. (29). Because $\chi_{+,+} = -\chi_{-,-}$, the linear canting term shifts the two allowed gaps in opposite directions. The quadratic coefficient ξ_{sp} is common to both channels and therefore shifts their average energy without changing their separation to this order. The exact expressions are used in the numerical results.

To compare the strengths of the different optical transitions, we use the dimensionless single-particle oscillator strength for the transition from valence band v to conduction band c , polarized along ν ,

$$f_{cv}^\nu(\mathbf{k}) = \frac{2m_0}{\hbar^2} \frac{|M_{cv}^\nu(\mathbf{k})|^2}{E_c(\mathbf{k}) - E_v(\mathbf{k})}, \quad \bar{f}_{cv}^\nu(\mathbf{k}) = \frac{f_{cv}^\nu(\mathbf{k})}{f_0}, \quad (37)$$

where m_0 is the free-electron mass and f_0 is the maximum b -polarized band-edge oscillator strength among the transitions shown in Fig. 6. Figure 6 shows the resulting y -polarized oscillator strengths at the Γ point.

The exact absence of opposite-SRLP transitions and of the a -polarized band-edge response follows from the symmetries and orbital content of the minimal Hamiltonian. Spin-dependent tunneling, lower-symmetry spin-orbit terms, or additional orbital sectors can relax these restrictions and generate weak optical weight in channels that are dark in the reduced model.

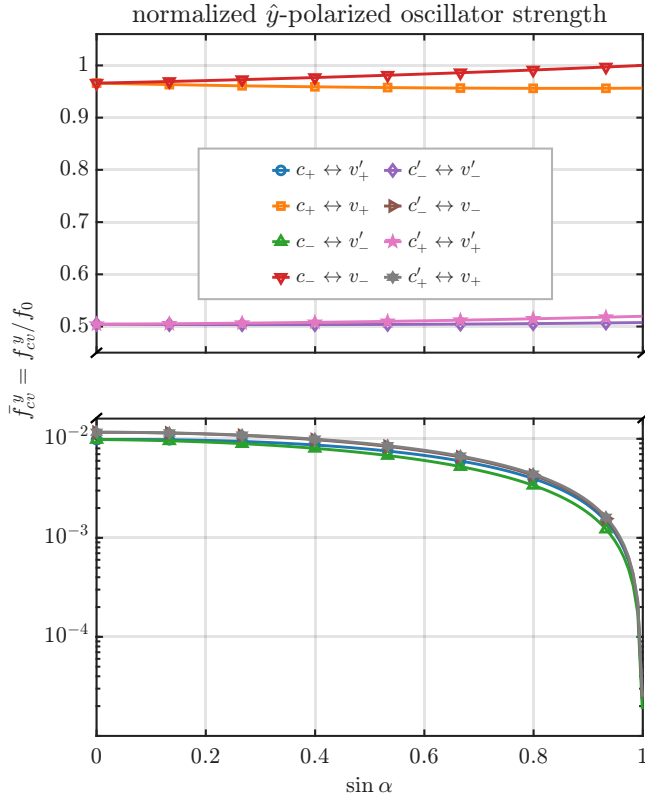


FIG. 6. Normalized b -polarized single-particle oscillator strengths $\bar{f}_{cv}^y = f_{cv}^y/f_0$ at the Γ point along the symmetric c -axis canting trajectory. The horizontal coordinate is $\mathcal{B} = \sin \alpha = B_{\text{ext}}/B_{\text{sat}}$, with the interlayer-AFM state at $\mathcal{B} = 0$ and the field-polarized interlayer-FM state at $\mathcal{B} = 1$. The upper panel shows the strong equal-SRLP transitions on a linear scale, whereas the lower panel shows the weaker equal-SRLP cross transitions on a logarithmic scale. The two lowest transitions $v_{\lambda} \rightarrow c_{\lambda}$ retain substantial oscillator strength throughout the canting trajectory. Equal-SRLP transitions involving an upper conduction band c'_{λ} or a lower valence band v'_{λ} are weaker and vanish at the FM state for the present parameterization, where spin and layer parity become separately conserved. Opposite-SRLP transitions are not shown because their oscillator strengths vanish exactly within the reduced Hamiltonian. The a -polarized band-edge oscillator strengths vanish because $\hat{V}_x(\Gamma) = 0$.

V. EXCITONS

We now include the screened electron-hole interaction in the field-dependent single-particle basis developed above. Within the Tamm-Dancoff approximation, a zero-center-of-mass ($\mathbf{Q} = 0$) exciton is a correlated electron-hole eigenstate expanded as a coherent superposition of vertical transitions from occupied valence bands to empty conduction bands [38–41]. Its amplitudes describe both the relative electron-hole motion and the distribution of the state among the retained band-pair channels.

Because the canted Bloch states are coherent mixtures of layer and spin components, a band-pair label does

not by itself determine whether the electron and hole occupy the same layer or different layers. We therefore formulate the Bethe-Salpeter equation in the band basis while retaining the layer-resolved matrix elements of the screened interaction.

A. Layer-resolved Bethe-Salpeter equation

Within the Tamm-Dancoff approximation, a normalized $\mathbf{Q} = 0$ electron-hole state is written as

$$|\Psi_n\rangle = \sum_{cv} \int \frac{d^2\mathbf{k}}{(2\pi)^2} \Psi_{n,cv}(\mathbf{k}) \hat{c}_{\mathbf{k}c}^{\dagger} \hat{c}_{\mathbf{k}v} |GS\rangle, \quad (38)$$

where $\hat{c}_{\mathbf{k}c}^{\dagger}$ creates an electron in conduction band c , while $\hat{c}_{\mathbf{k}v}$ removes an electron from valence band v . The exciton amplitudes satisfy

$$\sum_{cv} \int \frac{d^2\mathbf{k}}{(2\pi)^2} |\Psi_{n,cv}(\mathbf{k})|^2 = 1. \quad (39)$$

For compactness, we combine the conduction- and valence-band indices into an electron-hole channel index $a = (c, v)$ and write $\Psi_{n,a}(\mathbf{k}) \equiv \Psi_{n,cv}(\mathbf{k})$. Retaining the statically screened direct electron-hole attraction, the Bethe-Salpeter equation is

$$E_{n,X} \Psi_{n,a}(\mathbf{k}) = \Delta_a(\mathbf{k}) \Psi_{n,a}(\mathbf{k}) - \sum_b \int \frac{d^2\mathbf{k}'}{(2\pi)^2} W_{\text{RK}}(q) \mathcal{F}_{ab}(d; \mathbf{k}, \mathbf{k}') \Psi_{n,b}(\mathbf{k}'), \quad (40)$$

$$\Delta_a(\mathbf{k}) = E_c(\mathbf{k}) - E_v(\mathbf{k}),$$

Here, $b = (\bar{c}, \bar{v})$ labels the intermediate electron-hole channel and $q = |\mathbf{k} - \mathbf{k}'|$. The form factor $\mathcal{F}_{ab}(d; \mathbf{k}, \mathbf{k}')$ contains the corresponding layer-resolved Bloch-state overlaps and is specified below. The function $W_{\text{RK}}(q) > 0$ denotes the magnitude of the screened attraction.

We model the screened interaction using the Rytova-Keldysh form [42–44],

$$W_{\text{RK}}(q) = \frac{e^2}{2\epsilon_0 q (\epsilon_{\text{env}} + \rho_0 q)} = \frac{e^2}{2\epsilon_0 \epsilon_{\text{env}} q (1 + r_* q)}, \quad r_* = \frac{\rho_0}{\epsilon_{\text{env}}}. \quad (41)$$

Here, ρ_0 is the intrinsic two-dimensional screening length and r_* is the effective screening radius in an environment with dielectric constant ϵ_{env} . The conversion to the dimensionless reference units and the numerical screening parameters used below are given in Appendix B.

The form factor $\mathcal{F}_{ab}(d; \mathbf{k}, \mathbf{k}')$ accounts for the layer composition of the initial electron-hole channel a and the intermediate channel b . To resolve its same-layer and different-layer contributions, we use the one-particle layer projectors

$$\mathbb{P}_{\ell} = \Pi_{\ell} \otimes \tau_0 \otimes \sigma_0, \quad \Pi_{1,2} = \frac{\eta_0 \pm \eta_z}{2}, \quad \ell = 1, 2. \quad (42)$$

Here, $\mathbb{P}_\ell$ projects a Bloch state onto layer ℓ while leaving its orbital and spin components unchanged. Using these projectors, the layer-resolved form factor is separated into intralayer and interlayer contributions,

$$\mathcal{F}_{ab}(d; \mathbf{k}, \mathbf{k}') = \mathcal{F}_{ab}^{\text{intra}}(\mathbf{k}, \mathbf{k}') + e^{-dq} \mathcal{F}_{ab}^{\text{inter}}(\mathbf{k}, \mathbf{k}'). \quad (43)$$

For $a = (c, v)$ and $b = (\bar{c}, \bar{v})$, the corresponding matrix elements are

$$\mathcal{F}_{c\bar{c}v\bar{v}}^{\text{intra}}(\mathbf{k}, \mathbf{k}') = \sum_{\ell=1,2} \langle c\mathbf{k} | \mathbb{P}_\ell | \bar{c}\mathbf{k}' \rangle \langle \bar{v}\mathbf{k}' | \mathbb{P}_\ell | v\mathbf{k} \rangle, \quad (44)$$

$$\mathcal{F}_{c\bar{c}v\bar{v}}^{\text{inter}}(\mathbf{k}, \mathbf{k}') = \sum_{\ell \neq m} \langle c\mathbf{k} | \mathbb{P}_\ell | \bar{c}\mathbf{k}' \rangle \langle \bar{v}\mathbf{k}' | \mathbb{P}_m | v\mathbf{k} \rangle. \quad (45)$$

The intralayer contribution describes electron-hole configurations whose charge centers occupy the same layer, whereas the interlayer contribution describes configurations whose charge centers occupy different layers. The factor e^{-dq} accounts for the Fourier-space attenuation of the interaction between charge distributions separated by the effective distance d [45, 46].

When the intralayer screening is kept independent of d , varying d changes only the interlayer attraction through the factor e^{-dq} . The intralayer interaction and the single-particle Hamiltonian remain fixed. This dependence therefore describes the weakening of the interlayer attraction rather than the complete effect of changing the physical layer spacing. The case in which the second layer also screens the intralayer electron-hole interaction is discussed in Appendix G 1.

The in-plane screening of CrSBr can in general depend on the direction of the transferred momentum. Here, we retain the strongly anisotropic single-particle dispersion while approximating the interaction by the isotropic function $W_{\text{RK}}(q)$. An anisotropic interaction can be incorporated by replacing $W_{\text{RK}}(q)$ with $W(\mathbf{q})$ without changing the layer-resolved structure of Eqs. (44) and (45) [31].

The present BSE retains the screened direct interaction and neglects the electron-hole exchange kernel. The exact separation into equal- and opposite-SRLP sectors therefore refers to the direct-kernel minimal model. Long-range exchange can modify the fine structure and radiative properties without altering the single-particle SRLP classification [31]. Details of the momentum-space discretization, eigenvector normalization, treatment of the $q = 0$ singularity, and numerical convergence are given in Appendix G.

B. Symmetry reduction and two-channel exciton model

We restrict the layer-resolved Bethe-Salpeter equation to the two lowest conduction bands, c_+ and c_- , and the two highest valence bands, v_+ and v_- . These near-gap bands define four electron-hole channels,

$$(c_+v_-), \quad (c_-v_-), \quad (c_+v_+), \quad (c_-v_+). \quad (46)$$

The channel labels identify the conduction and valence bands forming the electron-hole pair. They do not specify its intralayer or interlayer character, because the conduction and valence states forming each electron-hole channel are themselves superpositions of the two layers.

Each channel carries the two-particle SRLP eigenvalue $\Lambda_{eh} = \lambda_c \lambda_v = \pm 1$. We denote by $W_{\ell m}(q)$ the screened direct interaction between an electron in layer ℓ and a hole in layer m . For equivalent layers, layer-exchange symmetry gives $W_{11}(q) = W_{22}(q)$ and $W_{12}(q) = W_{21}(q)$.

Under this condition it conserves Λ_{eh} , and channels belonging to different sectors do not mix. A layer-asymmetric dielectric environment or inequivalent electronic layers generally relax this exact decomposition. In the channel order of Eq. (46), the BSE Hamiltonian has the block structure

$$\mathcal{H}_{\mathbf{k}\mathbf{k}'}^{\text{BSE}} = \begin{pmatrix} D_{+, -} & 0 & 0 & -K_{+, -, -, +} \\ 0 & D_{-, -} & -K_{-, -, +, +} & 0 \\ 0 & -K_{+, +, -, -} & D_{+, +} & 0 \\ -K_{-, +, +, -} & 0 & 0 & D_{-, +} \end{pmatrix}_{\mathbf{k}\mathbf{k}'}. \quad (47)$$

These matrix elements are those of the BSE kernel in Eq. (40). For channels $a = (c_\lambda, v_\lambda)$ and $b = (c_{\lambda'}, v_{\lambda'})$, we define

$$K_{ab}(\mathbf{k}, \mathbf{k}') = W_{\text{RK}}(q) \mathcal{F}_{ab}(d; \mathbf{k}, \mathbf{k}'),$$

$$D_a(\mathbf{k}, \mathbf{k}') = (2\pi)^2 \delta^{(2)}(\mathbf{k} - \mathbf{k}') \Delta_a(\mathbf{k}) - K_{aa}(\mathbf{k}, \mathbf{k}'). \quad (48)$$

Thus, D_a is the intrachannel BSE kernel, whereas K_{ab} with $a \neq b$ is the direct-interaction matrix element that couples two different electron-hole channels. The symmetry reduction is derived in Appendix H.

The channels (c_-v_-) and (c_+v_+) have $\Lambda_{eh} = +1$ and form the equal-SRLP sector. The channels (c_+v_-) and (c_-v_+) have $\Lambda_{eh} = -1$ and form an independent dark sector within the reduced Hamiltonian. The opposite-SRLP channels are dark by the SRLP electric-dipole selection rule. An individual exciton in the equal-SRLP sector can nevertheless have weak or vanishing oscillator strength because of destructive interference between its channel amplitudes.

The binding energy is defined relative to the lowest independent electron-hole continuum in the same invariant sector,

$$E_{b,n}^{(\Lambda_{eh})}(\alpha) = E_{\text{th}}^{(\Lambda_{eh})}(\alpha) - E_{n,X}^{(\Lambda_{eh})}(\alpha), \quad (49)$$

$$E_{\text{th}}^{(\Lambda_{eh})}(\alpha) = \min_{\mathbf{k}, (c,v) \in \mathcal{C}_{\Lambda_{eh}}} [E_c(\mathbf{k}, \alpha) - E_v(\mathbf{k}, \alpha)], \quad (50)$$

where

$$\begin{aligned} \mathcal{C}_+ &= \{(c_-, v_-), (c_+, v_+)\}, \\ \mathcal{C}_- &= \{(c_+, v_-), (c_-, v_+)\}. \end{aligned} \quad (51)$$

A state is bound when $E_{b,n}^{(\Lambda_{eh})} > 0$. Because the equal- and opposite-SRLP sectors can have different continuum thresholds, their binding energies are evaluated separately.

We first consider the two lowest excitons in the equal-SRLP sector, formed from the bright single-particle transitions $v_- \rightarrow c_-$ and $v_+ \rightarrow c_+$. At the AFM state, if the

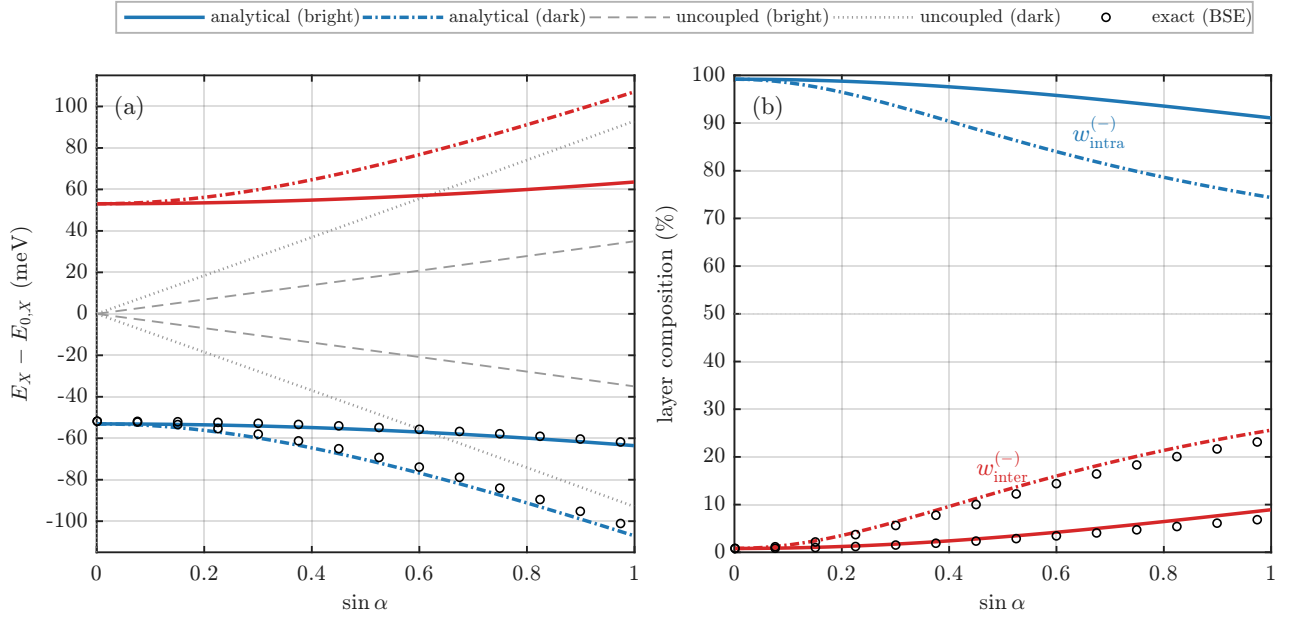


FIG. 7. Two-channel exciton description and layer composition of the lowest excitons at $d = 8.00 \text{ \AA}$. Solid and dash-dotted curves denote the bright equal-SRLP and dark opposite-SRLP ($\Lambda_{eh} = -1$) sectors. The horizontal coordinate is $\mathcal{B} = \sin \alpha = B_{\text{ext}}/B_{\text{sat}}$. (a) Exciton energies relative to $E_{0,X}$, the common energy at the AFM state when the coupling between the two excitons is omitted. The curves show the two-channel exciton models, the energies obtained by setting the coupling between the two excitons to zero in gray, and the full-BSE results as open circles. At the AFM state $\mathcal{B} = 0$, the two excitons in each SRLP sector are degenerate when their mutual coupling is omitted. Including the coupling gives the equal-SRLP energies $E_{0,X} - \Omega_+$ and $E_{0,X} + \Omega_+$, separated by $2\Omega_+$. In the opposite-SRLP sector, the analogous coupling Ω_- splits the two excitons by $2\Omega_-$. (b) Intralayer and interlayer probabilities of the lower-energy exciton in each SRLP sector. Each retained electron-hole channel contains both intralayer and interlayer configurations.

part of the direct electron-hole interaction that couples these two transitions is neglected, the corresponding exciton states are exactly degenerate. Magnetic canting shifts the energies of these two excitons in opposite directions.

To describe these two excitons analytically, we approximate their in-plane electron-hole wave functions as identical and retain their different band and layer-spin structure. Writing $\mathcal{B} = \sin \alpha$, the effective Hamiltonian for the two equal-SRLP excitons is

$$H_{\text{eff}}^{(+)}(\mathcal{B}) = (E_{0,X} + \xi_X \mathcal{B}^2) \mathbb{I}_2 + \begin{pmatrix} -\chi_+ \mathcal{B} & -\Omega_+ \\ -\Omega_+ & +\chi_+ \mathcal{B} \end{pmatrix}, \quad (52)$$

The two exciton branches are

$$E_{\pm,X}(\mathcal{B}) = E_{0,X} + \xi_X \mathcal{B}^2 \pm \sqrt{\Omega_+^2 + \chi_+^2 \mathcal{B}^2}. \quad (53)$$

where $E_{0,X}$ is the common energy of the two excitons at the AFM state when the coupling between them is set to zero, and $\xi_X \mathcal{B}^2$ is the quadratic energy shift common to both excitons. The parameters χ_+ and Ω_+ characterize, respectively, the canting-induced detuning and the coupling between the two equal-SRLP excitons. Their physical origins and explicit expressions are given below.

The opposite energy shifts under canting arise from the interplay of the exchange field and interlayer hopping in the conduction and valence bands. The corresponding

coefficient

$$\chi_+ = -\frac{J^A t_{\perp}^A}{R_A} - \frac{J^B t_{\perp}^B}{R_B} \quad (54)$$

is the leading canting-induced detuning. The linear relative shift of the two exciton energies therefore vanishes when the interlayer hopping is absent.

The coupling between the two equal-SRLP excitons has a different origin. Each of the two underlying SRLP transitions contains intralayer electron-hole configurations, in which the electron and hole occupy the same layer, and interlayer configurations, in which they occupy opposite layers. An interlayer electron-hole pair has a finite out-of-plane separation, and therefore a different Coulomb attraction from an intralayer pair. The two SRLP transitions combine these intralayer and interlayer configurations differently. Consequently, a difference between the intralayer and interlayer attraction energies couples the excitons formed from the two transitions and lifts their degeneracy at the AFM state.

The strength of this coupling also depends on the layer composition of the conduction and valence band-edge states. At the AFM state, this dependence is quantified by

$$Q_0 = |\langle c_+(\Gamma) | \eta_z | c_-(\Gamma) \rangle \langle v_+(\Gamma) | \eta_z | v_-(\Gamma) \rangle|_{\alpha=0}. \quad (55)$$

Here, $c_{\pm}(\Gamma)$ and $v_{\pm}(\Gamma)$ denote the conduction- and valence-band-edge states with SRLP eigenvalue $\lambda = \pm 1$, respectively. The notation $\cdots|_{\alpha=0}$ specifies that these band-edge states are evaluated in the AFM configuration, and η_z distinguishes the two layers. Within the present model, $Q_0 = 1$ in the absence of interlayer hopping and decreases as interlayer hopping mixes the layer character of the band-edge states.

We denote by $\mathcal{V}_{\text{intra}}(0)$ and $\mathcal{V}_{\text{inter}}(0)$ the intralayer and interlayer electron-hole attraction energies, respectively, at the AFM state, obtained by averaging the corresponding Coulomb interaction over the common in-plane electron-hole wave function used above. Their average contributes equally to the energies of the two excitons, whereas their difference couples the two equal-SRLP excitons. The resulting coupling energy is

$$\Omega_+ = Q_0 \frac{|\mathcal{V}_{\text{intra}}(0) - \mathcal{V}_{\text{inter}}(0)|}{2}. \quad (56)$$

The detailed expressions for Q_0 , $\mathcal{V}_{\text{intra}}$, and $\mathcal{V}_{\text{inter}}$ are derived in Appendix H. The resulting exciton energies are compared with the full BSE calculation in Fig. 7(a).

At the AFM state $\mathcal{B} = 0$, the two equal-SRLP excitons are degenerate when their mutual coupling is omitted. The coupling Ω_+ lifts this degeneracy and separates their energies by $2\Omega_+$.

The square-root term in Eq. (53) can be expanded in powers of $\mathcal{B}$ with expansion parameter $|\chi_+ \mathcal{B} / \Omega_+|$. When $\Omega_+ > |\chi_+|$, this parameter remains smaller than unity throughout the physical canting range $0 \leq \mathcal{B} \leq 1$. Retaining terms through second order gives

$$E_{-,X}(\mathcal{B}) = E_{0,X} - \Omega_+ + \left(\xi_X - \frac{\chi_+^2}{2\Omega_+} \right) \mathcal{B}^2 + O(\mathcal{B}^4). \quad (57)$$

The quadratic coefficient therefore contains the common contribution ξ_X and the additional term $-\chi_+^2/(2\Omega_+)$ generated by the coupling between the two equal-SRLP excitons.

The opposite-SRLP sector is described by an analogous two-channel Hamiltonian with its own detuning and coupling. This Hamiltonian is derived in Appendix H 5 and is compared with the full BSE results in Fig. 7(a). Its exciton energies, binding energies, and layer probabilities are included in the numerical results.

C. Layer composition

The physical layer composition of an exciton is defined using the two-particle projectors

$$\begin{aligned} \mathcal{P}_{\text{intra}} &= \mathbb{P}_1^{(e)} \mathbb{P}_1^{(h)} + \mathbb{P}_2^{(e)} \mathbb{P}_2^{(h)}, \\ \mathcal{P}_{\text{inter}} &= \mathbb{P}_1^{(e)} \mathbb{P}_2^{(h)} + \mathbb{P}_2^{(e)} \mathbb{P}_1^{(h)}, \end{aligned} \quad (58)$$

where $\mathbb{P}_{\ell}^{(e)}$ and $\mathbb{P}_{\ell}^{(h)}$ act on the electron and hole coordinates, respectively. Their expectation values give

$$\begin{aligned} w_{\text{intra}} &= \langle \Psi_n | \mathcal{P}_{\text{intra}} | \Psi_n \rangle, \quad w_{\text{inter}} = \langle \Psi_n | \mathcal{P}_{\text{inter}} | \Psi_n \rangle, \\ w_{\text{intra}} + w_{\text{inter}} &= 1. \end{aligned} \quad (59)$$

These quantities are physical intralayer and interlayer probabilities. They are not the squared coefficients of the $|c_- v_- \rangle$ and $|c_+ v_+ \rangle$ channel amplitudes, because each band-pair channel already contains both layer configurations.

Within the equal-SRLP effective Hamiltonian in Eq. (52), the interlayer probability of the lower exciton branch $E_{-,X}$ is

$$w_{\text{inter}}^{(-)}(\mathcal{B}) = \frac{1}{2} \left[1 - Q_0 \frac{\Omega_+}{\sqrt{\Omega_+^2 + \chi_+^2 \mathcal{B}^2}} \right], \quad (60)$$

with $w_{\text{intra}}^{(-)}(\mathcal{B}) = 1 - w_{\text{inter}}^{(-)}(\mathcal{B})$. The analytical layer probabilities are compared with the full BSE results in Fig. 7(b). The corresponding expression for the opposite-SRLP sector is given in Appendix H 5.

The same coupling Ω_+ that opens the avoided crossing also changes the intralayer and interlayer composition of the equal-SRLP exciton states. Defining the mean energy and half-splitting,

$$\begin{aligned} \bar{E}_X(\mathcal{B}) &= \frac{E_{+,X}(\mathcal{B}) + E_{-,X}(\mathcal{B})}{2} = E_{0,X} + \xi_X \mathcal{B}^2, \\ S_X(\mathcal{B}) &= \frac{E_{+,X}(\mathcal{B}) - E_{-,X}(\mathcal{B})}{2} = \sqrt{\Omega_+^2 + \chi_+^2 \mathcal{B}^2}, \end{aligned} \quad (61)$$

gives $S_X(0) = \Omega_+$. Equation (60) can therefore be written as

$$w_{\text{inter}}^{(-)}(\mathcal{B}) = \frac{1}{2} \left[1 - Q_0 \frac{S_X(0)}{S_X(\mathcal{B})} \right]. \quad (62)$$

Hence, if both equal-SRLP exciton branches are spectroscopically resolved, their measured separation $2S_X(\mathcal{B})$, together with the single-particle factor Q_0 obtained from the model, provides a model-based estimate of the intralayer and interlayer probabilities. This relation connects the field-dependent exciton spectrum to the evolution of its layer composition.

D. Optical response

Magneto-optical spectroscopy probes how the energies and spectral weights of exciton resonances evolve with the magnetic configuration. In CrSBr, these resonances have been studied using polarization-resolved reflectance, differential-reflectance, absorption, and photoluminescence measurements [4, 12, 18]. The present calculation describes the linear absorption response and therefore connects most directly to absorption and reflectance measurements. Photoluminescence intensities are not calculated because they additionally depend on exciton populations and relaxation processes.

Using the interband matrix element $M_{cv}^{\nu}(\mathbf{k})$ defined in Eq. (31), we define the single-particle dipole matrix element as

$$r_{cv}^{\nu}(\mathbf{k}) = -i \frac{M_{cv}^{\nu}(\mathbf{k})}{E_c(\mathbf{k}) - E_v(\mathbf{k})} = -i\hbar \frac{V_{cv}^{\nu}(\mathbf{k})}{E_c(\mathbf{k}) - E_v(\mathbf{k})}. \quad (63)$$

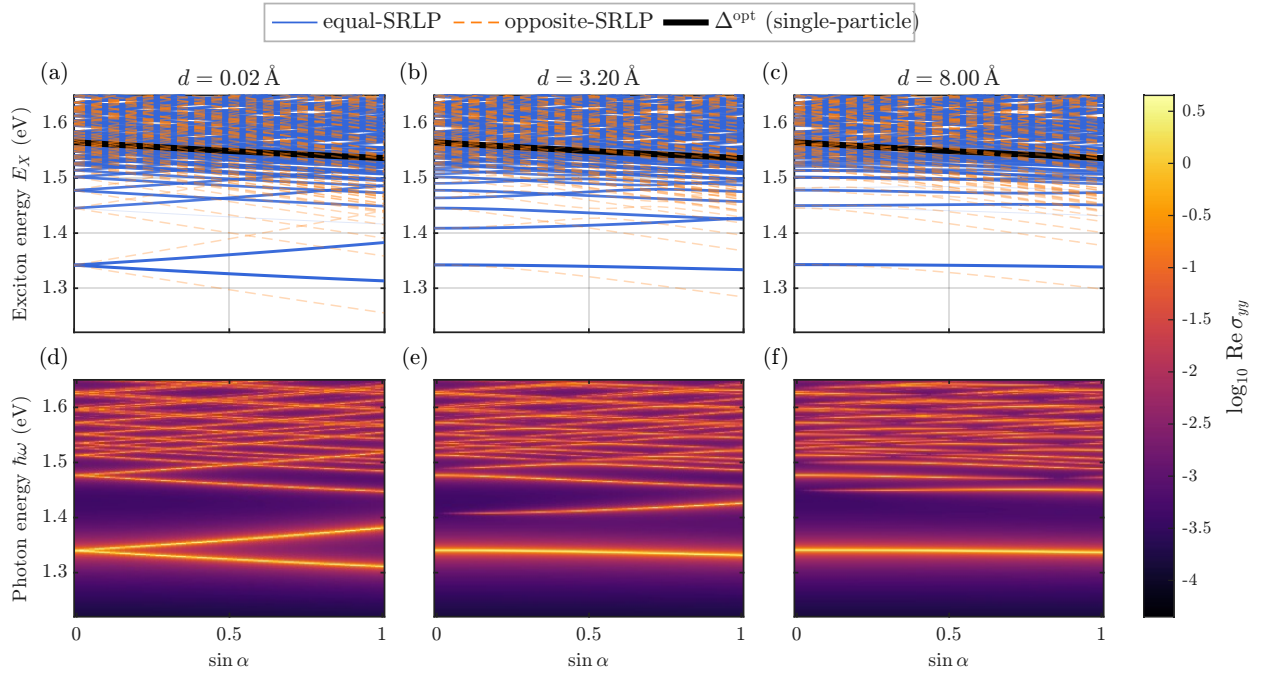


FIG. 8. Exciton spectrum and normalized b -polarized optical response along the symmetric c -axis canting trajectory for three effective charge-center separations. The horizontal coordinate is $\mathcal{B} = \sin \alpha = B_{\text{ext}}/B_{\text{sat}}$. Panels (a)-(c) show the exciton spectra, while panels (d)-(f) show the corresponding normalized optical response. The thick solid curves in panels (a)-(c) show the corresponding single-particle optical gaps $\Delta_{\chi}^{\text{opt}}$. (a,d) Nearly equal intralayer and interlayer attractions, $d = 0.02$ Å. (b,e) Intermediate separation, $d = 3.20$ Å. (c,f) CrSBr calculation, $d = 8.00$ Å. Increasing d attenuates only the interlayer attraction in this simulation and enlarges the avoided-crossing splitting. The quadratic dependence of the lower equal-SRLP exciton near the AFM state is resolved more directly in Fig. 7(a). The color scale in panels (d)-(f) is $\log_{10}[\text{Re } \tilde{\sigma}_{yy}]$ in normalized model units. A Lorentzian broadening of $\gamma = 0.3$ meV is used as a numerical resolution parameter.

For polarization along direction ν , the transition amplitude between the ground state and exciton n is

$$\begin{aligned} X_{n0}^{\nu} &= \sum_{cv} \int \frac{d^2\mathbf{k}}{(2\pi)^2} \Psi_{n,cv}^*(\mathbf{k}) r_{cv}^{\nu}(\mathbf{k}), \\ X_{0n}^{\nu} &= (X_{n0}^{\nu})^*, \end{aligned} \quad (64)$$

where the sum runs over the retained conduction-valence band pairs [37–39]. The sum is coherent, so contributions from different momenta and band-pair channels can interfere. Although the constituent equal-SRLP single-particle transitions are optically allowed, an exciton formed from them can have weak or vanishing optical weight because their contributions to X_{n0}^{ν} can interfere destructively. In contrast, excitons in the opposite-SRLP sector have zero electric-dipole amplitude within the reduced Hamiltonian.

The quantity $|X_{n0}^{\nu}|^2$ is the squared transition-dipole amplitude. In the response convention used below, the resonant spectral weight is proportional to $E_{n,X}|X_{n0}^{\nu}|^2$. The excitonic optical response is written as [37–39]

$$\begin{aligned} \tilde{\sigma}_{\nu\nu'}(\omega) &= -i \sum_n \frac{E_{n,X} X_{0n}^{\nu} X_{n0}^{\nu'}}{E_{n,X} - \hbar\omega - i\gamma} \\ &\quad + i \sum_n \frac{E_{n,X} X_{n0}^{\nu} X_{0n}^{\nu'}}{E_{n,X} + \hbar\omega + i\gamma}, \end{aligned} \quad (65)$$

where γ is a phenomenological linewidth used to broaden the discrete BSE eigenvalues. The poles occur at the exciton energies $E_{n,X}$, while their spectral weights are determined by the coherent optical amplitudes. The numerical spectra below show the b -polarized response $\text{Re } \tilde{\sigma}_{yy}$.

VI. EXCITONS: NUMERICAL RESULTS

A. Exciton spectrum and layer evolution

Figure 7 compares the two-channel exciton model with the full BSE. At the AFM state, the two-channel model gives $\Omega_+ = 53.1$ meV, compared with 51.3 meV from the full BSE. The layer factor and attraction energies entering the two-channel model are defined in Appendix H.

The origin of the splitting is isolated by removing the off-diagonal BSE matrix elements between $|c_-v_- \rangle$ and $|c_+v_+ \rangle$. The two equal-SRLP excitons are then degenerate at the AFM state within numerical precision. Restoring these BSE matrix elements opens the full 102.6 meV gap. The AFM splitting is therefore generated by level repulsion arising from the difference between the intralayer and interlayer electron-hole attractions rather than by a pre-existing detuning of the two excitons.

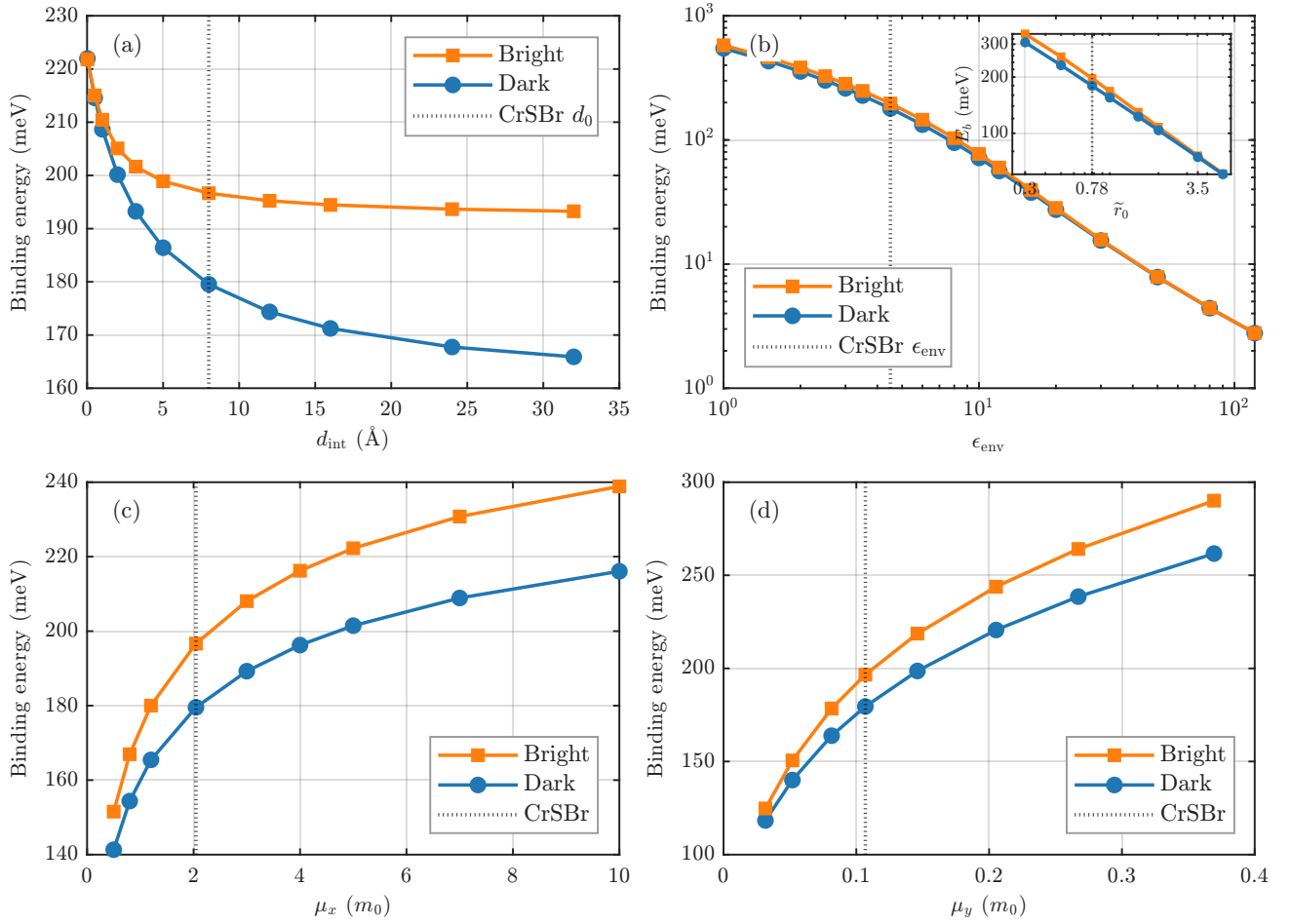


FIG. 9. Binding energies at the field-polarized FM state of the lowest equal-SRLP ($\Lambda_{eh} = +1$) and opposite-SRLP ($\Lambda_{eh} = -1$) excitons. All panels are obtained by solving the momentum-dependent layer-resolved BSE for the four near-gap electron-hole channels in Eq. (46), without the analytical two-channel exciton approximation. (a) Effective charge-center separation d when the modification of the intralayer screening by the other layer is neglected. Here $W_{\text{intra}} = W_{\text{RK}}$ and $W_{\text{inter}} = e^{-\tilde{d}q} W_{\text{RK}}$, so only the interlayer attraction changes. (b) Environmental dielectric constant ϵ_{env} . The main panel varies ϵ_{env} , which changes both the interaction strength and the effective Rytova-Keldysh screening radius. The inset instead varies the dimensionless effective screening radius $\tilde{r}_*$ at fixed interaction strength. The dotted lines mark the CrSBr values $\epsilon_{\text{env}} = 4.5$ and $\tilde{r}_* = 0.778$, respectively. (c) Reduced mass μ_x . Here μ_x is varied while μ_y and the interaction parameters are held fixed. (d) Reduced mass μ_y . Here μ_y is varied while μ_x and the interaction parameters are held fixed. All other parameters are fixed in each simulation. Dotted lines mark the reference parameter set.

The two-channel exciton model also reproduces the field dependence and physical layer composition of the lower branch. The rms differences from the full BSE are less than 0.1% for the exciton energy and 1% points for the interlayer probability.

For the reference parameter set, the interlayer probability of the lowest equal-SRLP exciton increases from 0.8% in the AFM state to 7.1% near saturation. The state remains predominantly intralayer, but its interlayer component grows substantially relative to its zero-field value.

Applying the same projection to the opposite-SRLP sector gives a two-channel Hamiltonian with different detuning and coupling parameters. It also reproduces the full-BSE opposite-SRLP branches and their layer

probabilities. Thus, the level-repulsion mechanism occurs in both SRLP sectors, although only excitons in the equal-SRLP sector can carry electric-dipole weight in the direct-kernel minimal model.

Figure 8 shows the calculated exciton spectrum and b -polarized optical response along the symmetric canting trajectory for three effective charge-center separations of the interaction model. Increasing the effective charge-center separation d suppresses the interlayer attraction through e^{-dq} while leaving the intralayer attraction unchanged. The resulting difference between the intralayer and interlayer attractions strengthens the coupling between the two equal-SRLP excitons and enlarges their avoided-crossing gap. The nearly coincident limit $d = 0.02$ Å represents almost equal intralayer and in-

terlayer attractions, while $d = 3.20 \text{ \AA}$ gives a moderate reduction of the interlayer attraction, whereas $d = 8.00 \text{ \AA}$ is used for the CrSBr calculation. In this comparison, d changes only the interlayer part of the electron-hole interaction and does not represent a physical change of the crystallographic layer spacing.

The bare orbital separation Δ_{Γ}^0 is chosen such that the lowest equal-SRLP exciton at the AFM state lies near 1.344 eV, within the energy range of the low-lying optical excitations reported for CrSBr [32, 33]. The same separation enters the relation between the band-edge masses, the interorbital matrix element, and the lattice hoppings in Appendix B. Changing Δ_{Γ}^0 while keeping the physical masses and p_{cv} fixed also changes the effective hoppings and can modify the exciton binding energy and wave function. A quantitative assignment to a particular experimental exciton therefore requires the band and electron-hole interaction parameters to be determined consistently for the same sample.

At the AFM state, the splitting between the two lowest equal-SRLP excitons increases from below 1 meV in the nearly coincident limit to about 103 meV at the reference separation [Figs. 8(a,c) and 7(a)]. For the same parameter set, the lower branch redshifts by approximately 4.3 meV between the AFM and field-polarized FM states. The nearly quadratic field dependence, shown more directly by the lower equal-SRLP branch in Fig. 7(a), and the simultaneous increase in interlayer character agree qualitatively with magneto-optical and GW-BSE results for CrSBr [18, 32].

The optical maps in Fig. 8 show both the resonance shifts and the transfer of spectral weight produced by coherent mixing of the two equal-SRLP excitons. They can be compared with relative peak positions and intensities in b -polarized absorption or magneto-reflectance.

B. Dependence of the binding energy on screening, separation, and mass anisotropy

We now examine how the interaction and anisotropic dispersion control the binding energy. To separate these effects from the magnetic evolution, we evaluate the lowest excitons at the field-polarized FM state and vary one parameter at a time. The binding energy is defined relative to the lowest independent electron-hole continuum in the same Λ_{eh} sector, as in Eq. (49). Numerical details are given in Appendix G.

Within the calculation that neglects the modification of intralayer screening by the other layer, increasing d weakens only the attraction between electron and hole configurations in different layers [Fig. 9(a)]. The binding therefore decreases but remains finite because the intralayer interaction is unchanged and the exciton becomes predominantly intralayer. A calculation that includes the modification of intralayer screening by the other layer also changes the screening of the intralayer interaction and gives the opposite trend. The binding increases with d

over most of the range considered because increasing the layer separation reduces the screening of the intralayer interaction by the other layer. This comparison is discussed in Appendix G 1 and Fig. 10.

Increasing ϵ_{env} screens both intralayer and interlayer attractions and reduces the binding of the exciton. The finite value at the largest permittivity in Fig. 9(b) reflects the finite simulation range; the binding approaches zero for asymptotically strong environmental screening.

The directional reduced masses satisfy

$$\frac{1}{\mu_{\nu}} = \frac{1}{m_{\nu}^e} + \frac{1}{m_{\nu}^h}, \quad \nu = x, y. \quad (66)$$

Increasing either reduced mass lowers the kinetic-energy cost of localizing the relative motion and increases the binding [Figs. 9(c) and 9(d)]. At the reference point, the FM equal-SRLP exciton logarithmic slopes are

$$\frac{d \ln E_b}{d \ln \mu_y} \approx 0.35, \quad \frac{d \ln E_b}{d \ln \mu_x} \approx 0.16. \quad (67)$$

The stronger sensitivity to μ_y follows from the small light-axis mass, which makes localization along b the larger kinetic-energy cost. Since the interaction is isotropic in this calculation, the directional response comes from the band dispersion rather than anisotropic screening. This trend is consistent with effective-mass treatments of CrSBr excitons [11, 31].

VII. DISCUSSION AND CONCLUSIONS

Magnetic order controls whether interlayer hopping is off-resonant or resonant. In the AFM state, the layer-staggered exchange field detunes same-spin states in opposite layers, and hopping shifts their energies only in second order. In the field-polarized FM state, the exchange field is layer uniform, so the corresponding states are resonant and acquire a first-order bonding-antibonding splitting. Canting continuously connects these limits. The wave functions change together with the energies. In particular, the conduction and valence states acquire a different layer composition as the moments cant, and these field-dependent electronic states form the basis of the exciton problem.

In the equal-SRLP sector, the excitons formed from the c_-v_- and c_+v_+ transitions are degenerate at the AFM state when the BSE matrix elements coupling these two transitions are omitted. Canting shifts their energies in opposite directions. Because each transition contains both intralayer and interlayer electron-hole configurations, the difference between the intralayer and interlayer electron-hole attraction energies couples the two excitons. The resulting avoided crossing produces the quadratic curvature of the lower exciton and changes its physical layer composition.

Previous experiments and GW-BSE calculations established magnetic control of interlayer hybridization and

exciton energies in CrSBr [4, 18]. The present model identifies the corresponding analytical structure. SRLP remains conserved along the symmetric canting trajectory, the Γ -point eigenstates and direct gaps can be obtained at arbitrary canting, and for equivalent layers the direct BSE separates into two Λ_{eh} sectors. This construction complements observations of magnetically confined excitons [10], magnetic-order control of Coulomb correlations [11], distinct Frenkel- and Wannier-like excitonic responses [12], and bulk and surface excitons [47].

If both equal-SRLP exciton branches are resolved, their AFM separation is $2\Omega_+$ within the two-channel exciton model, while their field-dependent separation tests the predicted detuning. The transfer of spectral weight provides a second signature of coherent mixing between the two excitons. Inferring layer probabilities additionally requires the calculated overlap factor Q_0 and is therefore not a model-independent extraction. Likewise, an optical line shift cannot be equated directly with a binding-energy change because the independent electron-hole continuum also moves with magnetic order. Resolving both branches would therefore provide information that is not contained in the shift of a single optical line.

The absolute binding energy is more sensitive to the screened electron-hole interaction than the avoided-crossing splitting. Replacing the interaction that keeps the intralayer screening independent of layer separation by one that includes the screening of one layer by the other changes the FM equal-SRLP binding energy by about 10% for the reference parameter set, whereas the AFM exciton splitting changes by about 3.5%. The dependence on layer separation is more sensitive. If the intralayer interaction is kept fixed, increasing the separation weakens the interlayer attraction and reduces the binding. When the screening of the intralayer interaction by the other layer is also included, increasing the separation reduces this additional screening and the binding increases over most of the range considered. Both interactions satisfy $W_{11} = W_{22}$ and $W_{12} = W_{21}$, so the direct BSE retains its two Λ_{eh} sectors. The avoided crossing, the quadratic variation near the AFM state, and the increase of interlayer character toward the field-polarized state also remain, although their numerical magnitudes depend on the screened interaction.

The exact single-particle SRLP result relies on the symmetric canting trajectory and the minimal hopping structure. Spin-dependent tunneling, additional spin-orbit terms, inequivalent layers, or lower-symmetry magnetic configurations can mix its sectors. At the exciton level, separation into the two Λ_{eh} sectors additionally requires equivalent layers and a layer-symmetric direct interaction. Layer-asymmetric screening can therefore mix sectors that are independent here, while electron-hole exchange and additional electronic bands can modify the exciton fine structure and optical weights.

The generalized s-d model is not limited to the exciton problem. For a specified magnetic configuration, the exchange and hopping terms determine the electronic

energies, interlayer hybridization, and spin and layer composition of the Bloch states. These states determine the optical matrix elements considered here and also enter electronic transport. Changes in magnetic order can therefore modify spin-dependent interlayer transport and magnetoresistance through the same electronic Hamiltonian, although an explicit transport calculation additionally requires the relevant contacts and scattering processes [20]. If the localized-spin operators are retained rather than replaced by their ordered expectation values, fluctuations of the magnetic moments couple directly to the electron spin density through $\hat{H}_{ex}$. Their expansion in magnon modes gives electron-magnon coupling, and after the electron-hole interaction is included the same exchange term can also couple excitons to magnons [14].

For another layered magnetic semiconductor, the orbital states, exchange couplings, interlayer hopping, and spin-orbit terms in Eq. (1) must be chosen for that material. Chromium trihalides such as CrBr₃ and CrI₃ provide examples in which these ingredients differ from those used for CrSBr [1–3]. Additional layers can be included by adding the corresponding layer blocks, while several magnetic ions in the unit cell require additional magnetic-sublattice indices. Recent excitonic spectroscopy of four- and five-layer CrSBr has resolved layer-dependent magnetic configurations during magnetic switching [48], providing a direct setting in which the multilayer form of Eq. (6) can be applied. Commensurate collinear, canted, and noncollinear magnetic orders can be represented by the corresponding ordered moment directions and magnetic unit cell. The resulting optical and excitonic selection rules must then be obtained from the symmetries of that Hamiltonian. In particular, the SRLP symmetry found here relies on the layer-spin structure of the reduced CrSBr Hamiltonian along the symmetric c -axis canting trajectory. In conclusion, we have developed an orbital-, layer-, and spin-resolved generalized s-d model to address the magnetically tunable electronic and excitonic phenomena in layered magnetic semiconductors and applied it to the magnetic-order-dependent electronic structure, optical selection rules, and Wannier-Mott excitons of a CrSBr bilayer.

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DATA AVAILABILITY

The numerical data underlying the figures in this article are openly available in Zenodo [49]. The code used to

generate and analyze the data is available from the authors upon reasonable request.

Appendix A: Hard-axis canting from a two-sublattice free energy

Let $\mathbf{m}_1$ and $\mathbf{m}_2$ be dimensionless unit vectors along the ordered layer moments, and define the physical magnetizations by $\mathbf{M}_\ell = M_s \mathbf{m}_\ell$. We use a common free-energy-density normalization for every term below and set $B_{\text{ext}} = \mu_0 H_{\text{ext}}$. In the SI volumetric convention, M_s is the saturation magnetization in A/m and F , $\mathcal{J}_{\text{AF}}$, and $K_{h,e}$ are in J/m³; an areal convention is equivalent after integrating all terms through the same layer thickness. We write [18, 50, 51]

$$F = \mathcal{J}_{\text{AF}} \mathbf{m}_1 \cdot \mathbf{m}_2 + K_h(m_{1z}^2 + m_{2z}^2) - K_e(m_{1y}^2 + m_{2y}^2) - M_s B_{\text{ext}}(m_{1z} + m_{2z}). \quad (\text{A1})$$

Here $\mathcal{J}_{\text{AF}} > 0$, $K_h > 0$, and $K_e > 0$ have the same energy-density units as F , while $M_s B_{\text{ext}}$ has those units in the chosen magnetic normalization. The notation $\mathcal{J}_{\text{AF}}$ distinguishes the magnetic interlayer exchange from the orbital-dependent electronic exchange couplings J^τ .

The symmetric canting configuration is

$$\begin{aligned} \mathbf{m}_1 &= \cos \alpha \hat{y} + \sin \alpha \hat{z}, \\ \mathbf{m}_2 &= -\cos \alpha \hat{y} + \sin \alpha \hat{z}. \end{aligned} \quad (\text{A2})$$

Substitution gives

$$F(\alpha) = -\mathcal{J}_{\text{AF}} \cos(2\alpha) + 2(K_h + K_e) \sin^2 \alpha - 2K_e - 2M_s B_{\text{ext}} \sin \alpha. \quad (\text{A3})$$

The minimization of free energy gives

$$\frac{\partial F}{\partial \alpha} = 2 \cos \alpha [2(\mathcal{J}_{\text{AF}} + K_h + K_e) \sin \alpha - M_s B_{\text{ext}}] = 0. \quad (\text{A4})$$

Equation (A4) has two solutions. One is $\alpha = \pi/2$, for which the two layer moments are parallel to the applied field. For $\cos \alpha \neq 0$, the other solution is the canted configuration

$$\sin \alpha = \frac{M_s B_{\text{ext}}}{2(\mathcal{J}_{\text{AF}} + K_h + K_e)}. \quad (\text{A5})$$

For the canted solution,

$$\frac{\partial^2 F}{\partial \alpha^2} = 4(\mathcal{J}_{\text{AF}} + K_h + K_e) \cos^2 \alpha > 0,$$

so this solution is a minimum as long as $\sin \alpha < 1$. For the parallel solution $\alpha = \pi/2$,

$$\left. \frac{\partial^2 F}{\partial \alpha^2} \right|_{\alpha=\pi/2} = 2[M_s B_{\text{ext}} - 2(\mathcal{J}_{\text{AF}} + K_h + K_e)].$$

The parallel state therefore becomes the stable solution when the applied field exceeds the value at which the canted solution reaches $\sin \alpha = 1$.

This field is the saturation field,

$$B_{\text{sat}} = \frac{2(\mathcal{J}_{\text{AF}} + K_h + K_e)}{M_s}. \quad (\text{A6})$$

Thus, for $0 \leq B_{\text{ext}} < B_{\text{sat}}$, the stable configuration is the canted state with $\sin \alpha = B_{\text{ext}}/B_{\text{sat}}$. At $B_{\text{ext}} = B_{\text{sat}}$, the moments become parallel to the applied field, and they remain parallel for $B_{\text{ext}} > B_{\text{sat}}$. This is the canting relation used in Eq. (12).

The parameters $\mathcal{J}_{\text{AF}}$, K_h , and K_e can be estimated from measurements of the interlayer magnetic exchange, magnetic anisotropy, and saturation field in CrSBr [5, 20, 50–52].

Appendix B: Low-energy units and parameter extraction

We obtain the low-energy lattice parameters by matching Eq. (11) to an anisotropic two-band $\mathbf{k} \cdot \mathbf{p}$ Hamiltonian near the Γ point [31, 33, 36]. In the orbital basis $\{A, B\}$, the canonical momentum convention is

$$H_{\mathbf{k}, \mathbf{p}}(\mathbf{k}) = \begin{pmatrix} \epsilon^A + \frac{\hbar^2 k_x^2}{2m_{A,x}} + \frac{\hbar^2 k_y^2}{2m_{A,y}} & \frac{\hbar k_y p_{cv}}{m_0} \\ \frac{\hbar k_y p_{cv}^*}{m_0} & \epsilon^B + \frac{\hbar^2 k_x^2}{2m_{B,x}} + \frac{\hbar^2 k_y^2}{2m_{B,y}} \end{pmatrix}, \quad (\text{B1})$$

where $p_{cv} = \langle A | \hat{p}_y | B \rangle$ is the canonical interband momentum matrix element and m_0 is the free-electron mass. The quantities $m_{A,\nu}$ and $m_{B,\nu}$ are the signed curvature masses of the uncoupled A and B orbital sectors. They are not identical to the physical electron and hole masses because the interorbital $k_y p_{cv}$ term also contributes to the band curvature along y .

We use the two-dimensional excitonic units of Ref. [31] at the fixed reference dielectric $\epsilon_{\text{ref}} = 4.5$,

$$\begin{aligned} a_B^{2D} &= \frac{4\pi\epsilon_0\epsilon_{\text{ref}}\hbar^2}{2\mu_{\text{ref}}e^2}, \\ Ry^{2D} &= \frac{2\mu_{\text{ref}}e^4}{(4\pi\epsilon_0\epsilon_{\text{ref}})^2\hbar^2} = \frac{\hbar^2}{2\mu_{\text{ref}}(a_B^{2D})^2}, \quad \mu_{\text{ref}} \equiv \mu_y. \end{aligned} \quad (\text{B2})$$

Here, a_B^{2D} and Ry^{2D} are the length and energy units used to nondimensionalize the exciton calculation. The reference mass $\mu_{\text{ref}} = \mu_y$ is the electron-hole reduced mass along $y \parallel b$, with μ_y defined explicitly in Eq. (B13). These units are fixed by the reference parameter set and are kept unchanged when a mass or dielectric parameter is varied below.

We use $\nu = x, y$ below for a spatial direction and reserve μ_x and μ_y for the electron-hole reduced masses.

The dimensionless variables are

$$\begin{aligned}\tilde{\mathbf{k}} &= a_B^{2D} \mathbf{k}, & \tilde{H} &= \frac{H}{Ry^{2D}}, \\ \tilde{m}_{\tau,\nu} &= \frac{m_{\tau,\nu}}{\mu_{\text{ref}}}, & \tilde{\epsilon}^\tau &= \frac{\epsilon^\tau}{Ry^{2D}}.\end{aligned}\quad (\text{B3})$$

The resulting dimensionless Hamiltonian is

$$\tilde{H}_{\mathbf{k},\mathbf{p}}(\tilde{\mathbf{k}}) = \begin{pmatrix} \tilde{\epsilon}^A + \frac{\tilde{k}_x^2}{\tilde{m}_{A,x}} + \frac{\tilde{k}_y^2}{\tilde{m}_{A,y}} & \tilde{k}_y \tilde{p}_{cv} \\ \tilde{k}_y \tilde{p}_{cv}^* & \tilde{\epsilon}^B + \frac{\tilde{k}_x^2}{\tilde{m}_{B,x}} + \frac{\tilde{k}_y^2}{\tilde{m}_{B,y}} \end{pmatrix}, \quad (\text{B4})$$

where

$$\tilde{p}_{cv} = \frac{\hbar p_{cv}}{m_0 a_B^{2D} Ry^{2D}} = 2 \frac{\mu_{\text{ref}}}{m_0} \frac{a_B^{2D} p_{cv}}{\hbar}. \quad (\text{B5})$$

Equation (B5) follows by matching $\hbar k_y p_{cv}/m_0$ to $Ry^{2D} \tilde{k}_y \tilde{p}_{cv}$.

Define the two diagonal matrix elements of Eq. (B4) by

$$\begin{aligned}h_A(\tilde{\mathbf{k}}) &= \tilde{\epsilon}^A + \frac{\tilde{k}_x^2}{\tilde{m}_{A,x}} + \frac{\tilde{k}_y^2}{\tilde{m}_{A,y}}, \\ h_B(\tilde{\mathbf{k}}) &= \tilde{\epsilon}^B + \frac{\tilde{k}_x^2}{\tilde{m}_{B,x}} + \frac{\tilde{k}_y^2}{\tilde{m}_{B,y}},\end{aligned}\quad (\text{B6})$$

and set

$$\bar{\epsilon}_{\mathbf{k}} = \frac{h_A + h_B}{2}, \quad \mathcal{M}_{\mathbf{k}} = \frac{h_A - h_B}{2}. \quad (\text{B7})$$

The relative phase of the A and B orbitals may be chosen so that $\tilde{p}_{cv}$ is real. In this gauge,

$$\tilde{H}_{\mathbf{k},\mathbf{p}} = \bar{\epsilon}_{\mathbf{k}} \tau_0 + \mathcal{M}_{\mathbf{k}} \tau_z + \tilde{k}_y \tilde{p}_{cv} \tau_x. \quad (\text{B8})$$

The phases of the two orbital basis states are arbitrary. If $p_{cv} = |p_{cv}|e^{i\phi}$, replacing $|B\rangle$ by $e^{-i\phi}|B\rangle$, with $|A\rangle$ unchanged, makes p_{cv} real. This change of basis does not alter the energy spectrum or the magnitudes of optical matrix elements. Every A-B matrix element acquires the same relative phase under this transformation, so the same orbital basis convention must be used for p_{cv} and g_y .

For a complex momentum matrix element, the final term is replaced by

$$\tilde{k}_y [\text{Re} \tilde{p}_{cv} \tau_x - \text{Im} \tilde{p}_{cv} \tau_y].$$

The eigenvalues of Eq. (B4) are

$$\tilde{E}_\beta(\tilde{\mathbf{k}}) = \bar{\epsilon}_{\mathbf{k}} + \beta \sqrt{\mathcal{M}_{\mathbf{k}}^2 + \tilde{k}_y^2 |\tilde{p}_{cv}|^2}, \quad \beta = \pm 1. \quad (\text{B9})$$

Here, $\beta = \pm 1$ labels the upper and lower eigenvalue branches and is distinct from the spatial-direction index $\nu = x, y$.

Expanding about Γ -point gives

$$\begin{aligned}\frac{1}{\tilde{m}_{c,x}} &= \frac{1}{\tilde{m}_{A,x}}, & \frac{1}{\tilde{m}_{c,y}} &= \frac{1}{\tilde{m}_{A,y}} + \frac{|\tilde{p}_{cv}|^2}{\tilde{\Delta}_\Gamma^0}, \\ \frac{1}{\tilde{m}_{v,x}} &= \frac{1}{\tilde{m}_{B,x}}, & \frac{1}{\tilde{m}_{v,y}} &= \frac{1}{\tilde{m}_{B,y}} - \frac{|\tilde{p}_{cv}|^2}{\tilde{\Delta}_\Gamma^0},\end{aligned}\quad (\text{B10})$$

where

$$\tilde{\Delta}_\Gamma^0 = \tilde{\epsilon}^A - \tilde{\epsilon}^B = \frac{\Delta_\Gamma^0}{Ry^{2D}}. \quad (\text{B11})$$

Let $m_{e,\nu} > 0$ and $m_{h,\nu} > 0$ denote the physical electron and hole masses. Since the signed valence-band curvature satisfies $m_{v,\nu} = -m_{h,\nu}$, the orbital masses entering the bare two-orbital model are

$$\begin{aligned}\tilde{m}_{A,x} &= \tilde{m}_{e,x}, & \tilde{m}_{A,y} &= \left[\tilde{m}_{e,y}^{-1} - \frac{|\tilde{p}_{cv}|^2}{\tilde{\Delta}_\Gamma^0} \right]^{-1}, \\ \tilde{m}_{B,x} &= -\tilde{m}_{h,x}, & \tilde{m}_{B,y} &= \left[-\tilde{m}_{h,y}^{-1} + \frac{|\tilde{p}_{cv}|^2}{\tilde{\Delta}_\Gamma^0} \right]^{-1}.\end{aligned}\quad (\text{B12})$$

The negative sign of $m_{B,\nu}$ describes the downward curvature of the valence-like B orbital. The physical hole masses $m_{h,\nu}$ remain positive.

The reduced masses governing the relative electron-hole motion are

$$\mu_x^{-1} = m_{e,x}^{-1} + m_{h,x}^{-1}, \quad \mu_y^{-1} = m_{e,y}^{-1} + m_{h,y}^{-1}. \quad (\text{B13})$$

The lattice dispersion in Eq. (11) satisfies

$$2t_\nu^\tau [1 - \cos(k_\nu a_\nu)] = t_\nu^\tau a_\nu^2 k_\nu^2 + O(k_\nu^4). \quad (\text{B14})$$

Defining

$$\tilde{a}_\nu = \frac{a_\nu}{a_B^{2D}}, \quad \tilde{t}_\nu^\tau = \frac{t_\nu^\tau}{Ry^{2D}},$$

comparison with Eq. (B4) gives

$$\tilde{t}_\nu^\tau = \frac{1}{\tilde{m}_{\tau,\nu} \tilde{a}_\nu^2}. \quad (\text{B15})$$

Thus, a positive curvature mass gives a positive hopping in the convention of Eq. (11), whereas the negative curvature of the valence-like B sector gives a negative hopping.

The momentum-odd interorbital coupling in the main text is

$$d_y(\mathbf{k}) = -g_y \sin(k_y a_y) \simeq -g_y a_y k_y.$$

With $\tilde{g}_y = g_y/Ry^{2D}$, matching its term linear in k_y to the $\tilde{k}_y \tilde{p}_{cv} \tau_x$ term in Eq. (B8) gives

$$\tilde{g}_y = -\frac{\tilde{p}_{cv}}{\tilde{a}_y}. \quad (\text{B16})$$

The minus sign in Eq. (B16) follows from matching $d_y(\mathbf{k}) \simeq -g_y a_y k_y$ to the $+k_y \tilde{p}_{cv} \tau_x$ term in Eq. (B8), using the same A-B orbital basis in both descriptions. Changing the relative phase of the two orbital basis states changes the phases of g_y and p_{cv} consistently and leaves Eq. (B16) invariant. After choosing both matrix elements to be real, the remaining sign change $|B\rangle \rightarrow -|B\rangle$ reverses the signs of both g_y and p_{cv} , so only their relative sign is fixed by the matching.

Finally, the orbital-preserving lattice dispersion in Eq. (11) can equivalently be written as

$$\epsilon_\tau(\mathbf{k}) = \epsilon^\tau - 2t_x^\tau \cos(k_x a_x) - 2t_y^\tau \cos(k_y a_y).$$

Therefore,

$$\epsilon^\tau = \bar{\epsilon}^\tau + 2t_x^\tau + 2t_y^\tau. \quad (\text{B17})$$

The parameters quoted in physical units in the main text and in Appendix D are obtained by converting these dimensionless quantities back to eV, Å, and m_0 .

Equations (B10)-(B17) determine the orbital curvature masses, lattice hoppings, and onsite energies from the band-edge masses, interband momentum matrix element, and direct gap. The quantities $m_{e,\nu}$, $m_{h,\nu}$, p_{cv} , and Δ_Γ^0 should be taken from the same band description because p_{cv} and Δ_Γ^0 also enter the y -direction curvature in Eq. (B10). The screening parameters separately set the electron-hole attraction used in the BSE and should correspond to the dielectric environment being modeled. Mixing band parameters or screening parameters obtained for different structures or dielectric environments can therefore change the calculated dispersion and exciton binding energy [31, 33, 35, 36].

Appendix C: Proof of spin-resolved layer-parity conservation

To verify SRLP conservation, write

$$H(\mathbf{k}) \equiv H_{\text{CrSBr}}(\mathbf{k})$$

and decompose Eq. (13) as

$$H(\mathbf{k}) = H_{\text{orb}}(\mathbf{k}) + H_z(\alpha) + H_y(\alpha) + H_\perp, \quad (\text{C1})$$

with

$$\begin{aligned} H_{\text{orb}}(\mathbf{k}) &= \eta_0 \otimes [(\epsilon_A(\mathbf{k})\tau_A + \epsilon_B(\mathbf{k})\tau_B + d_y(\mathbf{k})\tau_x) \otimes \sigma_0], \\ H_z(\alpha) &= -\eta_0 \otimes [J_{\text{orb}} \otimes \sin \alpha \sigma_z], \\ H_y(\alpha) &= -\eta_z \otimes [J_{\text{orb}} \otimes \cos \alpha \sigma_y], \\ H_\perp &= -\eta_x \otimes [\mathcal{T}_\perp \otimes \sigma_0], \end{aligned}$$

where

$$J_{\text{orb}} = J^A \tau_A + J^B \tau_B, \quad \mathcal{T}_\perp = t_\perp^A \tau_A + t_\perp^B \tau_B. \quad (\text{C2})$$

The SRLP operator is

$$\zeta = \eta_x \otimes \tau_0 \otimes \sigma_z. \quad (\text{C3})$$

The orbital term commutes with ζ because it is proportional to η_0 and σ_0 , while ζ is proportional to τ_0 in orbital space. The uniform exchange term H_z commutes with ζ because its layer and spin factors are η_0 and σ_z . The interlayer hopping term $H_\perp$ also commutes with ζ because its layer factor is η_x and it is proportional to σ_0 in spin space.

The only nontrivial contribution is H_y , which is proportional to $\eta_z \otimes J_{\text{orb}} \otimes \sigma_y$. Using

$$\eta_x \eta_z = -\eta_z \eta_x, \quad \sigma_z \sigma_y = -\sigma_y \sigma_z,$$

we obtain

$$\begin{aligned} \zeta (\eta_z \otimes J_{\text{orb}} \otimes \sigma_y) &= (\eta_x \eta_z) \otimes J_{\text{orb}} \otimes (\sigma_z \sigma_y) \\ &= (\eta_z \eta_x) \otimes J_{\text{orb}} \otimes (\sigma_y \sigma_z) \\ &= (\eta_z \otimes J_{\text{orb}} \otimes \sigma_y) \zeta. \end{aligned} \quad (\text{C4})$$

The anticommutations in layer and spin space cancel. Therefore,

$$[\zeta, H(\mathbf{k})] = 0 \quad (\text{C5})$$

for every $\mathbf{k}$ and every canting angle α .

This result applies to the CrSBr Hamiltonian in Eq. (13). If an additional term δH is included, the same SRLP quantum number remains conserved if and only if

$$[\zeta, \delta H] = 0,$$

because the Hamiltonian already satisfies $[\zeta, H_{\text{CrSBr}}] = 0$. Spin-dependent hopping, spin-flip tunneling, and additional spin-orbit terms do not have a universal SRLP character. Their layer and spin matrix structure must be examined separately to determine whether they commute with ζ [10, 12].

Appendix D: CrSBr material parameters and numerical conventions

Table I lists the parameters used in the CrSBr calculations. The masses, dielectric constant, screening length, and reference layer separation are taken from Ref. [31]. The exchange fields, interlayer hoppings, odd interorbital coupling, and bare orbital separation belong to the reduced Hamiltonian introduced in the main text. Their values depend on the electronic structure used to construct that Hamiltonian. The parameters entering the band Hamiltonian and the screened electron-hole interaction must therefore correspond to the same material geometry and dielectric environment. The length and energy units are defined in Appendix B and are kept fixed in all parameter simulations.

TABLE I. Parameters used in the CrSBr calculations. The Rytova-Keldysh screening radius is $r_* = \rho_0/\epsilon_{\text{env}}$. In the fixed reference units its dimensionless value is $\tilde{r}_* = r_*/a_B^{2D}$. For $\epsilon_{\text{ref}} = 4.5$, the values used here give $\tilde{r}_* = 0.778$ and $r_* = 0.87$ nm. The parameters entering the dielectric and mass simulations in Fig. 9 are varied from the values listed here.

Quantity	Symbol	Value	Status/use
Lattice constants	a_x, a_y, a_z	3.50, 4.75, 7.94 Å	crystallographic inputs
Electron masses	m_x^e, m_y^e	7.31, 0.14 m_0	literature values
Hole masses	m_x^h, m_y^h	2.84, 0.45 m_0	literature values
Reduced masses	μ_x, μ_y	2.045, 0.1068 m_0	derived
Reference units	a_B^{2D}, Ry^{2D}	1.12 nm, 286 meV	defined in Appendix B
Reference dielectric	ϵ_{ref}	4.5	dielectric environment
Effective RK radius	r_*	0.87 nm ($\tilde{r}_* = 0.778$)	screening radius
Reference layer separation	d	8.0 Å	interlayer interaction
Reference optical alignment	E_X^{AFM}	1.344 eV	chosen exciton energy
Aligned AFM continuum gap	$E_g^{\Gamma, \text{AFM}}$	1.56510 eV	resulting continuum edge
Conduction exchange	J^A	+0.260 eV	effective parameter
Valence exchange	J^B	-0.450 eV	effective parameter
Conduction interlayer hopping	$t_\perp^A$	+0.029 eV	effective parameter
Valence interlayer hopping	$t_\perp^B$	+0.064 eV	effective parameter
Odd b -axis coupling	g_y	0.350 eV	optical and band-curvature parameter

At the reference dielectric constant, the dimensionless screened interaction is

$$\tilde{W}_{\text{RK}}(\tilde{q}) = \frac{2\pi}{\tilde{q} + 0.778\tilde{q}^2}. \quad (\text{D1})$$

In the layer-resolved BSE, the intralayer part of the interaction is multiplied by $\mathcal{F}^{\text{intra}}$, while the interlayer part is multiplied by $e^{-\tilde{d}\tilde{q}}\mathcal{F}^{\text{inter}}$. Increasing d therefore weakens the attraction between an electron and a hole in different layers without changing the intralayer attraction in this calculation. This is the dependence on d used in the spectra and in Fig. 9(a).

For the exchange fields and interlayer hoppings listed in Table I,

$$R_A = 0.261612 \text{ eV}, \quad R_B = 0.454528 \text{ eV}. \quad (\text{D2})$$

They follow from

$$R_\tau = \sqrt{(J^\tau)^2 + (t_\perp^\tau)^2} \quad (\text{D3})$$

at the AFM state.

The energy-aligned bare orbital separation is

$$\Delta_\Gamma^0 = E_g^{\Gamma, \text{AFM}} + R_A + R_B = 2.28124 \text{ eV}. \quad (\text{D4})$$

It is chosen before the BSE solution so that the 41×41 , $d = 8$ Å AFM equal-SRLP exciton lies at 1.344 eV. The associated threshold binding is 221.14 meV.

The value of Δ_Γ^0 also enters the relation between the physical y -direction masses and the bare orbital masses in Eq. (B10). Changing Δ_Γ^0 while keeping the physical electron and hole masses fixed therefore changes the A - and B -orbital curvatures. For this reason, the optical energy

is aligned before the lattice Hamiltonian is constructed rather than shifted after solving the BSE.

Including the contribution of g_y to the y -direction band curvature, the orbital hoppings are

$$\begin{aligned} t_x^A &= 0.04255 \text{ eV}, & t_y^A &= 1.15247 \text{ eV}, \\ t_x^B &= -0.10951 \text{ eV}, & t_y^B &= -0.32155 \text{ eV}. \end{aligned} \quad (\text{D5})$$

Their signs follow the dispersion convention in Eq. (11). The minus sign multiplying $\eta_x \mathcal{T}_\perp$ in Eq. (13) fixes the sign convention for the interlayer hopping.

The four electron-hole channels are ordered as

$$(c_+v_-, c_-v_-, c_+v_+, c_-v_+). \quad (\text{D6})$$

The equal-SRLP block contains (c_-v_-) and (c_+v_+) . The opposite-SRLP block contains (c_+v_-) and (c_-v_+) .

At the AFM state, the states with SRLP eigenvalues $\lambda = +1$ and -1 are exactly degenerate within each band pair. A numerical diagonalization can therefore return any orthonormal combination within the degenerate subspace. For the numerical simulations, we label the AFM states by following the SRLP-resolved eigenstates continuously as $\alpha \rightarrow 0^+$. This fixes the labels used in the numerical BSE without changing the exact degeneracy at $\alpha = 0$. The analytical results at $\alpha = 0$ refer to the exact AFM Hamiltonian and to eigenstates with definite SRLP.

The physical layer probabilities are obtained from

$$\begin{aligned} \mathcal{P}_{\text{intra}} &= \mathbb{P}_1^{(e)} \mathbb{P}_1^{(h)} + \mathbb{P}_2^{(e)} \mathbb{P}_2^{(h)}, \\ \mathcal{P}_{\text{inter}} &= \mathbb{P}_1^{(e)} \mathbb{P}_2^{(h)} + \mathbb{P}_2^{(e)} \mathbb{P}_1^{(h)}, \end{aligned} \quad (\text{D7})$$

with

$$w_{\text{intra}} + w_{\text{inter}} = 1. \quad (\text{D8})$$

The expectation values of $\mathcal{P}_{\text{intra}}$ and $\mathcal{P}_{\text{inter}}$ give the probabilities that the electron and hole occupy the same layer or different layers. These probabilities are different from the equal- and opposite-SRLP labels used to separate the BSE.

Appendix E: SRLP-resolved Γ -point Hamiltonian and eigenstates

At the Γ point, the odd interorbital coupling vanishes, $d_y(\Gamma) = 0$, so the A and B orbital sectors decouple. For a fixed orbital $\tau = \text{A, B}$, Eq. (13) reduces to

$$H_\tau(\Gamma) = \bar{\epsilon}^\tau - J^\tau (\sin \alpha \sigma_z + \cos \alpha \eta_z \sigma_y) - t_\perp^\tau \eta_x. \quad (\text{E1})$$

1. Fixed-SRLP block

The layer-parity states and the two SRLP sectors are given in Eq. (16). Keeping the state order specified there, the Γ -point Hamiltonian for orbital $\tau = \text{A, B}$ and SRLP eigenvalue $\lambda = \pm 1$ is

$$H_{\tau\lambda}(\Gamma) = \begin{pmatrix} \bar{\epsilon}^\tau - \lambda t_\perp^\tau - J^\tau \sin \alpha & iJ^\tau \cos \alpha \\ -iJ^\tau \cos \alpha & \bar{\epsilon}^\tau + \lambda t_\perp^\tau + J^\tau \sin \alpha \end{pmatrix}. \quad (\text{E2})$$

The interlayer hopping has opposite signs for the two layer-parity states. The uniform z -directed exchange field also has opposite signs for the two basis states because their σ_z eigenvalues are opposite. The staggered y -directed exchange term changes both layer parity and spin and therefore couples the two states without changing their SRLP eigenvalue.

Diagonalizing Eq. (E2) gives

$$\begin{aligned} R_{\tau\lambda}(\alpha) &= \sqrt{(\lambda t_\perp^\tau + J^\tau \sin \alpha)^2 + (J^\tau \cos \alpha)^2} \\ &= \sqrt{(J^\tau)^2 + (t_\perp^\tau)^2 + 2\lambda J^\tau t_\perp^\tau \sin \alpha}. \end{aligned} \quad (\text{E3})$$

The first expression separates the two contributions to the level splitting. The term $\lambda t_\perp^\tau + J^\tau \sin \alpha$ contains the interlayer hopping and the layer-uniform component of the exchange field. The term $J^\tau \cos \alpha$ comes from the layer-staggered component of the exchange field.

The eigenvalues are

$$E_{\tau\lambda}^\beta(\Gamma) = \bar{\epsilon}^\tau + \beta R_{\tau\lambda}(\alpha), \quad \beta = \pm 1. \quad (\text{E4})$$

The two states in a given orbital and SRLP sector are therefore separated by $2R_{\tau\lambda}$.

2. Normalized eigenstates at arbitrary canting

For $R_{\tau\lambda} \neq 0$, the relative weights of the two states in each SRLP sector can be written in terms of

$$\theta_{\tau\lambda} = \text{atan2}(J^\tau \cos \alpha, \lambda t_\perp^\tau + J^\tau \sin \alpha), \quad -\pi < \theta_{\tau\lambda} \leq \pi. \quad (\text{E5})$$

This gives

$$\cos \theta_{\tau\lambda} = \frac{\lambda t_\perp^\tau + J^\tau \sin \alpha}{R_{\tau\lambda}}, \quad \sin \theta_{\tau\lambda} = \frac{J^\tau \cos \alpha}{R_{\tau\lambda}}. \quad (\text{E6})$$

For $\lambda = +1$, the normalized conduction-band states are

$$|c_+\rangle = \cos\left(\frac{\theta_{A+}}{2}\right) |p_+, \text{A}, \uparrow_z\rangle + i \sin\left(\frac{\theta_{A+}}{2}\right) |p_-, \text{A}, \downarrow_z\rangle. \quad (\text{E7})$$

$$|c'_+\rangle = i \sin\left(\frac{\theta_{A+}}{2}\right) |p_+, \text{A}, \uparrow_z\rangle + \cos\left(\frac{\theta_{A+}}{2}\right) |p_-, \text{A}, \downarrow_z\rangle. \quad (\text{E8})$$

The normalized valence-band states are

$$|v'_+\rangle = \cos\left(\frac{\theta_{B+}}{2}\right) |p_+, \text{B}, \uparrow_z\rangle + i \sin\left(\frac{\theta_{B+}}{2}\right) |p_-, \text{B}, \downarrow_z\rangle. \quad (\text{E9})$$

$$|v_+\rangle = i \sin\left(\frac{\theta_{B+}}{2}\right) |p_+, \text{B}, \uparrow_z\rangle + \cos\left(\frac{\theta_{B+}}{2}\right) |p_-, \text{B}, \downarrow_z\rangle. \quad (\text{E10})$$

For $\lambda = -1$, the normalized conduction-band states are

$$|c_-\rangle = \cos\left(\frac{\theta_{A-}}{2}\right) |p_-, \text{A}, \uparrow_z\rangle + i \sin\left(\frac{\theta_{A-}}{2}\right) |p_+, \text{A}, \downarrow_z\rangle. \quad (\text{E11})$$

$$|c'_-\rangle = i \sin\left(\frac{\theta_{A-}}{2}\right) |p_-, \text{A}, \uparrow_z\rangle + \cos\left(\frac{\theta_{A-}}{2}\right) |p_+, \text{A}, \downarrow_z\rangle. \quad (\text{E12})$$

The normalized valence-band states are

$$|v'_-\rangle = \cos\left(\frac{\theta_{B-}}{2}\right) |p_-, \text{B}, \uparrow_z\rangle + i \sin\left(\frac{\theta_{B-}}{2}\right) |p_+, \text{B}, \downarrow_z\rangle. \quad (\text{E13})$$

$$|v_-\rangle = i \sin\left(\frac{\theta_{B-}}{2}\right) |p_-, \text{B}, \uparrow_z\rangle + \cos\left(\frac{\theta_{B-}}{2}\right) |p_+, \text{B}, \downarrow_z\rangle. \quad (\text{E14})$$

The coefficients $\cos^2(\theta_{\tau\lambda}/2)$ and $\sin^2(\theta_{\tau\lambda}/2)$ give the probabilities of the two spin and layer-parity basis states within the corresponding SRLP sector. Their variation with α describes the change of the single-particle wave functions during magnetic canting.

The two states with the same orbital character and SRLP eigenvalue become degenerate when $R_{\tau\lambda} = 0$. For $J^\tau \neq 0$ and $0 \leq \alpha \leq \pi/2$, Eq. (E3) gives

$$\alpha = \frac{\pi}{2}, \quad t_\perp^\tau = -\lambda J^\tau. \quad (\text{E15})$$

The CrSBr parameters used here do not satisfy this condition. At $R_{\tau\lambda} = 0$, the two states are degenerate and $\theta_{\tau\lambda}$ is undefined.

3. Antiferromagnetic limit

At $\alpha = 0$,

$$R_{\tau\lambda} = R_\tau \equiv \sqrt{(J^\tau)^2 + (t_\perp^\tau)^2} \quad (\text{E16})$$

is independent of λ . The two SRLP sectors are therefore exactly degenerate.

This degeneracy is enforced by the additional unitary operator

$$\mathcal{U}_{\text{AFM}} = \eta_x \otimes \tau_0 \otimes \sigma_x, \quad (\text{E17})$$

which satisfies

$$[\mathcal{U}_{\text{AFM}}, H(\mathbf{k}, \alpha = 0)] = 0, \quad \{\mathcal{U}_{\text{AFM}}, \zeta\} = 0. \quad (\text{E18})$$

Consequently, $\mathcal{U}_{\text{AFM}}$ maps every eigenstate with SRLP eigenvalue λ to a degenerate eigenstate with SRLP eigenvalue $-\lambda$.

At $\alpha = 0$, σ_y is conserved. For a fixed eigenvalue $s_y = \pm 1$,

$$H_{\tau, s_y}^{\text{AFM}} = \bar{\epsilon}^\tau - s_y J^\tau \eta_z - t_\perp^\tau \eta_x. \quad (\text{E19})$$

The two eigenvalues are

$$E_{\tau, \pm}^{\text{AFM}} = \bar{\epsilon}^\tau \pm R_\tau. \quad (\text{E20})$$

For $|t_\perp^\tau| \ll |J^\tau|$,

$$E_{\tau, \pm}^{\text{AFM}} = \bar{\epsilon}^\tau \pm |J^\tau| \pm \frac{(t_\perp^\tau)^2}{2|J^\tau|} + O\left[\frac{(t_\perp^\tau)^4}{|J^\tau|^3}\right]. \quad (\text{E21})$$

At $t_\perp^\tau = 0$, the layer states satisfy

$$E_{1, s_y}^{(0)} = \bar{\epsilon}^\tau - s_y J^\tau, \quad E_{2, s_y}^{(0)} = \bar{\epsilon}^\tau + s_y J^\tau. \quad (\text{E22})$$

For a fixed spin s_y , the states in layers 1 and 2 are therefore separated in energy by $2|J^\tau|$ before interlayer hopping is included.

Their leading wave-function corrections are

$$\begin{aligned} |1, s_y\rangle &\longrightarrow |1, s_y\rangle + \frac{t_\perp^\tau}{2s_y J^\tau} |2, s_y\rangle, \\ |2, s_y\rangle &\longrightarrow |2, s_y\rangle - \frac{t_\perp^\tau}{2s_y J^\tau} |1, s_y\rangle. \end{aligned} \quad (\text{E23})$$

Thus, a state localized in one layer at $t_\perp^\tau = 0$ acquires a component in the other layer already at first order in $t_\perp^\tau/J^\tau$. Its energy changes only at second order in the interlayer hopping, with a correction of order $(t_\perp^\tau)^2/|J^\tau|$.

4. Ferromagnetic limit

At $\alpha = \pi/2$,

$$H_\tau^{\text{FM}}(\Gamma) = \bar{\epsilon}^\tau - J^\tau \sigma_z - t_\perp^\tau \eta_x. \quad (\text{E24})$$

The operators σ_z and η_x commute separately with the Hamiltonian. The eigenstates can therefore be labeled by the spin eigenvalue $s = \pm 1$ and the layer-parity eigenvalue $p = \pm 1$,

$$|\tau, s, p\rangle = |p\rangle_\eta \otimes |\tau\rangle_\tau \otimes |s\rangle_\sigma. \quad (\text{E25})$$

The SRLP eigenvalue is

$$\lambda = sp. \quad (\text{E26})$$

The corresponding energies are

$$E_{\tau, s, p}^{\text{FM}} = \bar{\epsilon}^\tau - sJ^\tau - pt_\perp^\tau. \quad (\text{E27})$$

Equivalently, since $p = \lambda s$,

$$E_{\tau, s, \lambda}^{\text{FM}} = \bar{\epsilon}^\tau - sJ^\tau - \lambda s t_\perp^\tau. \quad (\text{E28})$$

For a fixed spin s , the exchange field shifts the states in layers 1 and 2 by the same energy $-sJ^\tau$. The two layer states are therefore degenerate before interlayer hopping is included. Interlayer hopping mixes these degenerate states into the layer-parity combinations $|p_+\rangle$ and $|p_-\rangle$.

The energies of the two layer-parity states differ by

$$|E_{\tau, s, p=+1}^{\text{FM}} - E_{\tau, s, p=-1}^{\text{FM}}| = 2|t_\perp^\tau|. \quad (\text{E29})$$

Interlayer hopping therefore changes the FM energies already to first order in $t_\perp^\tau$. In the AFM state, the same-spin states in layers 1 and 2 are separated by $2|J^\tau|$ before hopping is included, and their leading energy shift is instead second order in $t_\perp^\tau$.

5. Continuous labeling across the canting trajectory

For $0 < \alpha < \pi/2$, the SRLP eigenvalue λ remains exact, while the spin and layer composition changes continuously with the mixing angle in Eq. (E5). The labels c_λ , c'_λ , v_λ , and v'_λ are assigned using the energy-ordering convention in Eq. (17).

At the AFM state, the $\lambda = \pm 1$ sectors become degenerate but remain distinguishable as eigenstates of ζ . At the FM state, spin and layer parity are separately conserved and the SRLP eigenvalue becomes $\lambda = sp$. The same SRLP eigenvalue therefore identifies each band-edge state throughout the canted regime even though spin and layer parity are not separately conserved for $0 < \alpha < \pi/2$.

6. Expansion in the interlayer mixing parameter

Writing $\mathcal{B} = \sin \alpha$, the exact SRLP-dependent splitting can be expressed as

$$R_{\tau\lambda}(\mathcal{B}) = R_\tau \sqrt{1 + \lambda r_\tau \mathcal{B}}, \quad r_\tau = \frac{2J^\tau t_\perp^\tau}{R_\tau^2}. \quad (\text{E30})$$

Since $|\mathcal{B}| \leq 1$, the condition $|r_\tau| < 1$ ensures $|r_\tau \mathcal{B}| < 1$ throughout the canting range. The binomial expansion of Eq. (E30) gives

$$\begin{aligned} R_{\tau\lambda}(\mathcal{B}) &= R_\tau \left[1 + \frac{1}{2} \lambda r_\tau \mathcal{B} - \frac{1}{8} r_\tau^2 \mathcal{B}^2 + O(|r_\tau \mathcal{B}|^3) \right] \\ &= R_\tau + \lambda \frac{J^\tau t_\perp^\tau}{R_\tau} \mathcal{B} - \frac{(J^\tau t_\perp^\tau)^2}{2R_\tau^3} \mathcal{B}^2 + O(R_\tau |r_\tau \mathcal{B}|^3). \end{aligned} \quad (\text{E31})$$

Substitution into Eq. (26) gives Eq. (29) and the coefficients in Eq. (30). For $\max_\tau |r_\tau| \ll 1$, the expansion remains controlled throughout the canting range.

Appendix F: Analytical band-edge optical matrix elements

The momentum derivative of the Bloch Hamiltonian preserves SRLP symmetry because the SRLP operator is independent of crystal momentum. This fixes which interband matrix elements can remain finite before their polarization dependence is evaluated.

1. SRLP selection rule

Because the SRLP operator is momentum independent, the commutation $[\zeta, H_{\text{CrSBr}}(\mathbf{k})] = 0$ implies

$$[\zeta, \partial_{k_\nu} H_{\text{CrSBr}}(\mathbf{k})] = \partial_{k_\nu} [\zeta, H_{\text{CrSBr}}(\mathbf{k})] = 0. \quad (\text{F1})$$

For conduction and valence states with SRLP eigenvalues λ_c and λ_v ,

$$\zeta |c_{\lambda_c}\rangle = \lambda_c |c_{\lambda_c}\rangle, \quad \zeta |v_{\lambda_v}\rangle = \lambda_v |v_{\lambda_v}\rangle. \quad (\text{F2})$$

It follows that

$$\begin{aligned} \lambda_c M_{c_{\lambda_c} v_{\lambda_v}}^\nu &= \langle c_{\lambda_c} | \zeta \partial_{k_\nu} H_{\text{CrSBr}} | v_{\lambda_v} \rangle \\ &= \langle c_{\lambda_c} | \partial_{k_\nu} H_{\text{CrSBr}} \zeta | v_{\lambda_v} \rangle \\ &= \lambda_v M_{c_{\lambda_c} v_{\lambda_v}}^\nu. \end{aligned} \quad (\text{F3})$$

Therefore,

$$(\lambda_c - \lambda_v) M_{c_{\lambda_c} v_{\lambda_v}}^\nu(\mathbf{k}) = 0. \quad (\text{F4})$$

For opposite SRLP,

$$M_{c_{\lambda_c} v_{\lambda_v}}^\nu(\mathbf{k}) = 0 \quad \text{when} \quad \lambda_c \neq \lambda_v. \quad (\text{F5})$$

Opposite-SRLP conduction and valence states therefore have zero interband matrix element for either optical polarization. Equal SRLP is necessary for a finite matrix element but does not by itself guarantee one. The polarization dependence is determined separately by the orbital structure of $\partial_{k_\nu} H_{\text{CrSBr}}$.

2. Exact matrix elements at arbitrary canting

At the Γ point, Eq. (32) gives

$$\begin{aligned} \partial_{k_y} H_{\text{CrSBr}}(\Gamma) &= -a_y g_y \eta_0 \otimes \tau_x \otimes \sigma_0, \\ \partial_{k_x} H_{\text{CrSBr}}(\Gamma) &= 0. \end{aligned} \quad (\text{F6})$$

The y -polarized operator changes the orbital character $A \leftrightarrow B$ and is independent of layer and spin. Its matrix elements are therefore determined by the overlap of the layer and spin parts of the conduction and valence states.

For states with the same SRLP eigenvalue, the relative mixing angle between the B - and A -orbital states is

$$\delta\theta_\lambda = \theta_{B\lambda} - \theta_{A\lambda}. \quad (\text{F7})$$

For equal SRLP, direct evaluation gives

$$\begin{aligned} M_{c_\lambda v_\lambda}^y(\Gamma) &= -ia_y g_y \sin\left(\frac{\delta\theta_\lambda}{2}\right), \\ M_{c'_\lambda v'_\lambda}^y(\Gamma) &= -ia_y g_y \sin\left(\frac{\delta\theta_\lambda}{2}\right), \\ M_{c'_\lambda v_\lambda}^y(\Gamma) &= -a_y g_y \cos\left(\frac{\delta\theta_\lambda}{2}\right), \\ M_{c_\lambda v'_\lambda}^y(\Gamma) &= -a_y g_y \cos\left(\frac{\delta\theta_\lambda}{2}\right). \end{aligned} \quad (\text{F8})$$

The relative angle is fixed by the exchange fields, interlayer hoppings, and canting angle. Its cosine is

$$\cos \delta\theta_\lambda = \frac{J^A J^B + t_\perp^A t_\perp^B + \lambda \sin \alpha (J^A t_\perp^B + J^B t_\perp^A)}{R_{A\lambda} R_{B\lambda}}. \quad (\text{F9})$$

Its sine is

$$\sin \delta\theta_\lambda = \frac{\lambda \cos \alpha (J^B t_\perp^A - J^A t_\perp^B)}{R_{A\lambda} R_{B\lambda}}. \quad (\text{F10})$$

The phases of the matrix elements depend on the phase convention chosen for the individual Bloch states. Their absolute values are gauge independent,

$$\begin{aligned} |M_{c_\lambda v_\lambda}^y|^2 &= |M_{c'_\lambda v'_\lambda}^y|^2 = a_y^2 g_y^2 \sin^2\left(\frac{\delta\theta_\lambda}{2}\right), \\ |M_{c'_\lambda v_\lambda}^y|^2 &= |M_{c_\lambda v'_\lambda}^y|^2 = a_y^2 g_y^2 \cos^2\left(\frac{\delta\theta_\lambda}{2}\right). \end{aligned} \quad (\text{F11})$$

All opposite-SRLP matrix elements vanish by Eq. (F5). All x -polarized band-edge matrix elements vanish because $\partial_{k_x} H_{\text{CrSBr}}(\Gamma) = 0$.

The equal-SRLP matrix elements satisfy

$$\begin{aligned} |M_{c_\lambda v_\lambda}^y|^2 + |M_{c'_\lambda v_\lambda}^y|^2 &= a_y^2 g_y^2, \\ |M_{c_\lambda v'_\lambda}^y|^2 + |M_{c'_\lambda v'_\lambda}^y|^2 &= a_y^2 g_y^2. \end{aligned} \quad (\text{F12})$$

Canting transfers y -polarized matrix-element weight between c_λ and c'_λ for a fixed valence state while their sum remains $a_y^2 g_y^2$. Opposite-SRLP transitions remain forbidden. The oscillator strength also contains the transition-energy factor $1/(E_c - E_v)$ and therefore changes accordingly.

3. Antiferromagnetic and ferromagnetic limits

In the AFM state, $\alpha = 0$, and the mixing angles in Eq. (E6) satisfy

$$\cos \theta_{\tau\lambda} = \frac{\lambda t_\perp^\tau}{R_\tau}, \quad \sin \theta_{\tau\lambda} = \frac{J^\tau}{R_\tau}, \quad R_\tau = \sqrt{(J^\tau)^2 + (t_\perp^\tau)^2}. \quad (\text{F13})$$

The relative angle between the A and B-orbital states then obeys

$$\cos \delta\theta_\lambda = \frac{t_\perp^A t_\perp^B + J^A J^B}{R_{A\lambda} R_{B\lambda}}. \quad (\text{F14})$$

This quantity is independent of the sign of λ . The squared y -polarized matrix elements are

$$\begin{aligned} |M_{c_\lambda v_\lambda}^y|^2 &= |M_{c'_\lambda v'_\lambda}^y|^2 = \frac{a_y^2 g_y^2}{2} \left[1 - \frac{t_\perp^A t_\perp^B + J^A J^B}{R_A R_B} \right], \\ |M_{c'_\lambda v_\lambda}^y|^2 &= |M_{c_\lambda v'_\lambda}^y|^2 = \frac{a_y^2 g_y^2}{2} \left[1 + \frac{t_\perp^A t_\perp^B + J^A J^B}{R_A R_B} \right]. \end{aligned} \quad (\text{F15})$$

For the CrSBr parameters, $J^A J^B < 0$ and the exchange energies are larger than the interlayer hoppings. The transitions $c_\lambda \leftrightarrow v_\lambda$ and $c'_\lambda \leftrightarrow v'_\lambda$ consequently have larger matrix elements than $c'_\lambda \leftrightarrow v_\lambda$ and $c_\lambda \leftrightarrow v'_\lambda$.

In the FM state, $\alpha = \pi/2$, and

$$R_{\tau\lambda} = |t_\perp^\tau + \lambda J^\tau|. \quad (\text{F16})$$

Equation (E6) then gives

$$\cos \theta_{\tau\lambda} = \lambda \operatorname{sgn}(t_\perp^\tau + \lambda J^\tau), \quad \sin \theta_{\tau\lambda} = 0. \quad (\text{F17})$$

The squared matrix elements become

$$\begin{aligned} |M_{c_\lambda v_\lambda}^y|^2 &= |M_{c'_\lambda v'_\lambda}^y|^2 \\ &= \frac{a_y^2 g_y^2}{2} [1 - \operatorname{sgn}(t_\perp^A + \lambda J^A) \operatorname{sgn}(t_\perp^B + \lambda J^B)]. \end{aligned} \quad (\text{F18})$$

The other two equal-SRLP transitions satisfy

$$\begin{aligned} |M_{c'_\lambda v_\lambda}^y|^2 &= |M_{c_\lambda v'_\lambda}^y|^2 \\ &= \frac{a_y^2 g_y^2}{2} [1 + \operatorname{sgn}(t_\perp^A + \lambda J^A) \operatorname{sgn}(t_\perp^B + \lambda J^B)]. \end{aligned} \quad (\text{F19})$$

For the CrSBr parameters,

$$\operatorname{sgn}(t_\perp^A + \lambda J^A) \operatorname{sgn}(t_\perp^B + \lambda J^B) = -1 \quad (\lambda = \pm 1). \quad (\text{F20})$$

Hence

$$\begin{aligned} |M_{c_\lambda v_\lambda}^y(\Gamma)| &= |M_{c'_\lambda v'_\lambda}^y(\Gamma)| = a_y |g_y|, \\ M_{c'_\lambda v_\lambda}^y(\Gamma) &= M_{c_\lambda v'_\lambda}^y(\Gamma) = 0. \end{aligned} \quad (\text{F21})$$

The vanishing of the two cross transitions in the FM state follows from the separate conservation of spin and layer parity. It is therefore stronger than the SRLP selection rule alone.

In the AFM state, the matrix-element magnitudes are identical for $\lambda = +1$ and $\lambda = -1$ because $\cos \delta \theta_\lambda$ is independent of λ . For the CrSBr parameters, the same equality holds in the FM state because Eq. (F20) has the same value for both SRLP sectors.

Appendix G: Optical response and numerical implementation details

The calculation samples the full rectangular Brillouin zone of the effective lattice Hamiltonian on an odd cen-

tered grid,

$$\begin{aligned} k_{x,i} &= \frac{2\pi i}{a_x N_x}, & i &= -\frac{N_x-1}{2}, \dots, \frac{N_x-1}{2}, \\ k_{y,j} &= \frac{2\pi j}{a_y N_y}, & j &= -\frac{N_y-1}{2}, \dots, \frac{N_y-1}{2}. \end{aligned} \quad (\text{G1})$$

Direct Euclidean differences $|\mathbf{k}_m - \mathbf{k}_{m'}|$ are used in the effective continuum interaction. For the 41×41 grid, the largest sampled magnitudes are $k_{x,\max} = 0.87571 \text{ \AA}^{-1}$ and $k_{y,\max} = 0.64526 \text{ \AA}^{-1}$.

The use of a continuum interaction with direct Euclidean momentum differences is appropriate only when the exciton is concentrated near Γ and has negligible weight at the grid boundary. The calculation should therefore be interpreted as a Γ -centered effective-model discretization rather than a periodic lattice Coulomb kernel over the full zone.

For a uniform grid,

$$w_m = \frac{\Delta k_x \Delta k_y}{(2\pi)^2}, \quad \tilde{\Psi}_{n,ma} = \sqrt{w_m} \Psi_{n,a}(\mathbf{k}_m), \quad (\text{G2})$$

so $\sum_{m,a} |\tilde{\Psi}_{n,ma}|^2 = 1$. In this weighted basis, the Hermitian BSE matrix is

$$\begin{aligned} [\mathcal{H}^{\text{BSE}}]_{ma,m'b} &= \Delta_a(\mathbf{k}_m) \delta_{mm'} \delta_{ab} \\ &\quad - \sqrt{w_m w_{m'}} W_{\text{RK}}(q_{mm'}) \mathcal{F}_{ab}(d; \mathbf{k}_m, \mathbf{k}_{m'}), \end{aligned} \quad (\text{G3})$$

where $q_{mm'} = |\mathbf{k}_m - \mathbf{k}_{m'}|$.

For the dimensionless interaction factor $1/(\tilde{q} + \tilde{r}_* \tilde{q}^2)$, its uniform-grid prefactor is

$$\frac{\Delta \tilde{k}_x \Delta \tilde{k}_y}{2\pi}. \quad (\text{G4})$$

The singular element is evaluated by averaging the same RK interaction over the rectangular central momentum cell.

$$\bar{W}_0 = \frac{1}{\Delta \tilde{k}_x \Delta \tilde{k}_y} \int_{\mathcal{C}_0} d^2 \tilde{q} \tilde{W}_{\text{RK}}(|\tilde{q}|), \quad (\text{G5})$$

where $\mathcal{C}_0 = [-\Delta \tilde{k}_x/2, \Delta \tilde{k}_x/2] \times [-\Delta \tilde{k}_y/2, \Delta \tilde{k}_y/2]$.

Figure 8 uses a 51×51 grid, 61 canting points, and 800 BSE eigenstates per SRLP sector. Figure 7 uses 41×41 sampling. The one-parameter simulations in Fig. 9 also use 41×41 sampling, so the reported reference bindings correspond to the same grid.

An input $\alpha = 0$ to the canting calculation is evaluated at the regularized value $\alpha_{\text{reg}} = 10^{-7}$. Varying α_{reg} from 10^{-4} to 10^{-10} changes the AFM energies by less than 10^{-10} eV. The regularization only selects the states continuously connected to positive canting and does not act as a physical symmetry-breaking field.

For the AFM equal-SRLP state at $d = 8 \text{ \AA}$, the binding energy changes with the momentum-grid size as

$N_x = N_y$	21	25	31	35	41
E_b (meV)	218.431	218.495	219.548	220.253	221.140

The $35 \rightarrow 41$ change is 0.89 meV. At the FM state the lowest equal-SRLP binding changes only from 195.26 meV at $N = 21$ to 196.66 meV at $N = 41$. The binding energies are therefore stable to within a few meV. The residual upward drift is limited by the k_y sampling of the quasi-one-dimensional exciton, while the sensitivity trends are robust.

1. Screening of the intralayer interaction by the second layer

In Sec. V, the intralayer attraction is the monolayer Rytova-Keldysh potential $W_{\text{intra}}(q) = W_{\text{RK}}(q)$, and the interlayer attraction is $W_{\text{inter}}(q) = e^{-\tilde{d}q} W_{\text{RK}}(q)$. With these interactions the interlayer separation d enters only the interlayer term, so increasing d weakens the attraction between electron and hole configurations in different layers while leaving the intralayer attraction unchanged [Eq. (43)].

The two-layer Rytova-Keldysh interaction of Semina *et al.* [their Eqs. (9a,b)] [31] includes the electrostatic

screening produced by both layers. With $\xi = e^{-\tilde{d}q}$ and $r_1 = r_2 = \tilde{r}_*$,

$$W_{11}(q) = \frac{2\pi}{\tilde{q}} \frac{1 + \tilde{q}r_2(1 - \xi^2)}{D}, \quad W_{12}(q) = \frac{2\pi}{\tilde{q}} \frac{\xi}{D},$$

$$D = (1 + \tilde{q}r_1)(1 + \tilde{q}r_2) - \tilde{q}^2 r_1 r_2 \xi^2. \quad (\text{G6})$$

Here, $W_{\text{intra}} = W_{11}$ and $W_{\text{inter}} = W_{12}$. The two screening treatments enter the BSE through the *identical* layer-resolved form factors [Eqs. (44) and (45)]. Their momentum-dependent interaction coefficients differ.

At $d \rightarrow 0$, the two-layer interaction reduces to a single sheet with the combined screening length $r_1 + r_2$, with denominator $1 + (r_1 + r_2)\tilde{q}$. The interaction that neglects screening by the second layer retains the single-layer radius $\tilde{r}_*$, with denominator $1 + \tilde{r}_*\tilde{q}$. The two interactions therefore do *not* coincide at small d .

They approach the same isolated-monolayer limit only as $d \rightarrow \infty$, with $W_{11} \rightarrow W_{\text{RK}}(\tilde{r}_*)$ and $W_{12} \rightarrow 0$. At intermediate d , the ξ^2 term in D accounts for the screening of an intralayer electron-hole pair by the second layer. Consequently, W_{11} changes with d , whereas $W_{\text{intra}} = W_{\text{RK}}$ remains independent of d when this additional screening is neglected.

At the CrSBr reference ($d = 8$ Å, 41×41 grid), the two screening treatments give

Quantity	fixed intralayer screening	two-layer screening
FM $\Lambda_{eh} = +1$ binding	196.7	176.9 meV
AFM splitting $2\Omega_+$	102.6	99.1 meV
AFM→FM redshift	4.37	3.27 meV
Interlayer weight, AFM→FM	0.8 → 7.1%	0.8 → 5.4%

The avoided crossing, the quadratic AFM redshift, and the increase of interlayer character from the AFM to the FM state are all preserved. Including the screening produced by the second layer lowers the reference FM binding by approximately 20 meV, or 10%, while the AFM splitting changes by approximately 3.6 meV, or 3.5%. The spectra of Figs. 8 and 7, and the screening- and mass-dependence panels Fig. 9(b) to (d), are therefore qualitatively unchanged.

The one qualitative difference is in the dependence on the interlayer separation [Fig. 9(a)]. The single-particle energies, Bloch states, and equal-SRLP continuum threshold $E_{\text{th}}^{(+)} = 1.53624$ eV are the same in both cases. The opposite dependence on d therefore comes from the screened electron-hole interaction.

In the calculation that neglects the modification of intralayer screening by the second layer, increasing d modifies only the interlayer interaction through $e^{-\tilde{d}q}$, while the intralayer interaction is held fixed. The FM binding therefore decreases monotonically from 222 to 193 meV.

The decrease follows from the weakening of the attraction between an electron and a hole in different layers while the same-layer attraction remains unchanged.

In the two-layer screening calculation, increasing d simultaneously weakens the interlayer attraction and reduces the screening of the intralayer interaction by the second layer. For the lowest exciton, the latter effect dominates beyond an extremely small- d range. The two-layer binding is nearly flat and weakly nonmonotonic at very small d , with a 0.06 meV dip between $d = 0$ and 0.1 Å, and then increases toward the isolated-layer limit from 146 to 189 meV. The opposite-SRLP exciton has the same broad dependence on d .

The two screening treatments agree only as $d \rightarrow \infty$, not as $d \rightarrow 0$. Because the single-particle threshold is d -independent, the exciton energy $E_X = E_{\text{th}} - E_b$ changes in the opposite direction to the binding energy. The hopping amplitudes, exchange energies, effective masses, and Bloch states remain fixed as d changes. A physical change of the layer spacing can also modify these electronic quantities.

Figure 10 compares the two binding curves directly.

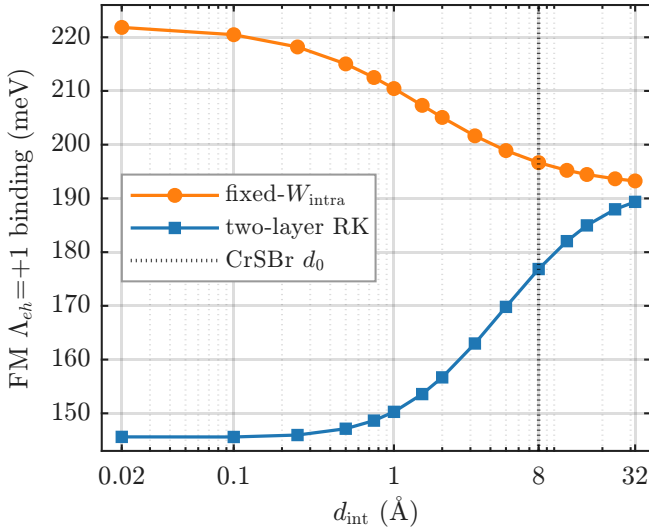


FIG. 10. FM binding energy of the lowest equal-SRLP ($\Lambda_{eh} = +1$) exciton as a function of the electron-hole layer separation d . The circular data points correspond to $W_{\text{intra}} = W_{\text{RK}}$ and $W_{\text{inter}} = e^{-d\tilde{q}}W_{\text{RK}}$, for which the screening of the intralayer interaction by the second layer is neglected. The square data points use the two-layer interaction in Eq. (G6), which includes this additional screening. When the intralayer interaction is kept fixed, the binding decreases as the interlayer attraction weakens with increasing d . With two-layer screening, the binding increases over most of the range because increasing d reduces the screening of the intralayer attraction by the second layer. Both interactions approach the isolated-layer limit at large d . The dotted line marks the CrSBr value $d_0 = 8$ Å.

With the velocity convention of Sec. IV,

$$r_{cv}^\alpha(\mathbf{k}) = -i \frac{\langle c, \mathbf{k} | \partial_{k_\alpha} H | v, \mathbf{k} \rangle}{E_c(\mathbf{k}) - E_v(\mathbf{k})}. \quad (\text{G7})$$

For Euclidean-normalized eigenvectors, the discrete optical amplitude on the momentum grid is

$$X_{n0}^\alpha = \sum_{m,a} \sqrt{w_m} \tilde{\Psi}_{n,ma}^* r_a^\alpha(\mathbf{k}_m), \quad X_{0n}^\alpha = (X_{n0}^\alpha)^*. \quad (\text{G8})$$

The normalized response uses Eq. (65) with $\gamma = 0.3$ meV. The spectra are normalized, so the overall optical-conductivity prefactor is omitted. For each SRLP sector, the exciton binding energy is measured from the lowest energy of an unbound electron-hole pair in that sector, as in Eqs. (49) and (50).

Appendix H: Two-particle symmetry and projection of the exciton Hamiltonian

For equivalent layers, the direct electron-hole interaction conserves the two-particle SRLP eigenvalue and separates the four near-gap electron-hole channels into two independent sectors. The two-channel exciton Hamiltonian in Sec. VB follows after projection onto the two

band-pair states within each sector. The symmetry decomposition is exact for the direct interaction retained in the manuscript. The analytical coupling and layer-probability formulas require the additional projection assumptions stated below.

1. Two-particle SRLP selection rule

The single-particle operator satisfies

$$\zeta^2 = \mathbb{I}, \quad [\zeta, H_{\text{CrSBr}}(\mathbf{k})] = 0. \quad (\text{H1})$$

For a one-particle operator O , the hole-space representation is $O^{(h)} = O^*$, so that

$$\langle h_v | O^{(h)} | h_{v'} \rangle = \langle v' | O | v \rangle. \quad (\text{H2})$$

The corresponding two-particle SRLP operator is

$$\mathcal{Z}_{eh} = \zeta^{(e)} \otimes \zeta^{(h)}, \quad \zeta^{(h)} = \zeta^*. \quad (\text{H3})$$

Because the SRLP eigenvalues are real,

$$\mathcal{Z}_{eh} | c_{\lambda_c} v_{\lambda_v} \rangle = \lambda_c \lambda_v | c_{\lambda_c} v_{\lambda_v} \rangle. \quad (\text{H4})$$

Thus, the two-particle eigenvalue is

$$\Lambda_{eh} = \lambda_c \lambda_v. \quad (\text{H5})$$

Conjugation by $\zeta = \eta_x \sigma_z$ exchanges the two one-particle layer projectors of Eq. (42),

$$\zeta \mathbb{P}_1 \zeta = \mathbb{P}_2, \quad \zeta \mathbb{P}_2 \zeta = \mathbb{P}_1. \quad (\text{H6})$$

It follows that the physical intralayer and interlayer projectors obey

$$\begin{aligned} \mathcal{Z}_{eh} \mathcal{P}_{\text{intra}} \mathcal{Z}_{eh}^{-1} &= \mathcal{P}_{\text{intra}}, \\ \mathcal{Z}_{eh} \mathcal{P}_{\text{inter}} \mathcal{Z}_{eh}^{-1} &= \mathcal{P}_{\text{inter}}. \end{aligned} \quad (\text{H7})$$

For a general layer-resolved direct interaction, this invariance requires

$$W_{11}(q) = W_{22}(q), \quad W_{12}(q) = W_{21}(q). \quad (\text{H8})$$

The quantity $W_{\ell m}(q)$ is the screened direct interaction between an electron in layer ℓ and a hole in layer m . Before projection onto the Bloch states, the layer dependence of the attractive direct interaction can be written as

$$K^{\text{att}}(q) = \sum_{\ell, m=1,2} W_{\ell m}(q) \mathbb{P}_\ell^{(e)} \mathbb{P}_m^{(h)}. \quad (\text{H9})$$

Its matrix elements give the layer-resolved direct-interaction term in Eq. (40).

The interaction forms in Sec. V and Appendix G 1 both satisfy Eq. (H8) for equivalent layers. The screened direct interaction does not act on spin. Therefore,

$$[\mathcal{Z}_{eh}, K^{\text{att}}] = 0. \quad (\text{H10})$$

This gives

$$K_{\lambda_c \lambda_v, \lambda'_c \lambda'_v}^{\text{att}} = 0 \quad \text{unless} \quad \lambda_c \lambda_v = \lambda'_c \lambda'_v. \quad (\text{H11})$$

The direct interaction therefore couples only electron-hole states with the same value of Λ_{eh} , giving the block structure in Eq. (47). If the two layers experience different intralayer screening, the pair of same-layer terms is no longer invariant under layer exchange and the two Λ_{eh} sectors can mix. Any additional interaction preserves the decomposition only if it commutes with $\mathcal{Z}_{eh}$.

2. Layer-projector matrix elements in the SRLP basis

Because η_z reverses layer parity without acting on spin, it connects the $\lambda = +1$ and $\lambda = -1$ states with the same orbital character. For the conduction states,

$$Q_A(\alpha) = |\langle c_+ | \eta_z | c_- \rangle|. \quad (\text{H12})$$

For the valence states,

$$Q_B(\alpha) = |\langle v_+ | \eta_z | v_- \rangle|. \quad (\text{H13})$$

For either orbital $\tau = A, B$, direct evaluation gives

$$Q_\tau^2(\alpha) = \frac{1}{2} \left[1 + \frac{(J^\tau)^2 - (t_\perp^\tau)^2}{R_{\tau,+}(\alpha) R_{\tau,-}(\alpha)} \right]. \quad (\text{H14})$$

At the AFM state,

$$Q_\tau(0) = \frac{|J^\tau|}{\sqrt{(J^\tau)^2 + (t_\perp^\tau)^2}}. \quad (\text{H15})$$

For $t_\perp^\tau = 0$, $Q_\tau(0) = 1$. Finite interlayer hopping reduces $Q_\tau(0)$ because the AFM eigenstates are no longer confined to a single layer.

The relative phases of the two channel states can be chosen so that the matrix elements of η_z are real and positive. Their product is

$$Q(\alpha) = Q_A(\alpha) Q_B(\alpha). \quad (\text{H16})$$

In the ordered channel basis

$$\{|c_- v_- \rangle, |c_+ v_+ \rangle\}, \quad (\text{H17})$$

projection of the physical layer projectors gives

$$\begin{aligned} \mathcal{P}_{\text{intra}}^{\text{proj}} &= \frac{1}{2} \begin{pmatrix} 1 & Q \\ Q & 1 \end{pmatrix}, \\ \mathcal{P}_{\text{inter}}^{\text{proj}} &= \frac{1}{2} \begin{pmatrix} 1 & -Q \\ -Q & 1 \end{pmatrix}. \end{aligned} \quad (\text{H18})$$

Their sum is $\mathbb{I}_2$. For $Q < 1$, the projected matrices are not idempotent because the two retained band-pair states do not span the full space of physical electron-hole layer configurations. Their expectation values nevertheless give the intralayer and interlayer probabilities within the retained two-channel subspace.

3. Attraction energies and coupling between equal-SRLP excitons

Let $\varphi(\mathbf{k}; \alpha)$ be the common normalized relative-motion envelope of the two equal-SRLP excitons,

$$\int \frac{d^2 \mathbf{k}}{(2\pi)^2} |\varphi(\mathbf{k}; \alpha)|^2 = 1. \quad (\text{H19})$$

The layer-even physical configurations are

$$\begin{aligned} |X_{\text{intra}}\rangle &= \frac{|1_e 1_h\rangle + |2_e 2_h\rangle}{\sqrt{2}}, \\ |X_{\text{inter}}\rangle &= \frac{|1_e 2_h\rangle + |2_e 1_h\rangle}{\sqrt{2}}. \end{aligned} \quad (\text{H20})$$

The corresponding positive electron-hole attraction energies are

$$\begin{aligned} \mathcal{V}_{\text{intra}}(\alpha) &= \langle \varphi X_{\text{intra}} | K^{\text{att}} | \varphi X_{\text{intra}} \rangle, \\ \mathcal{V}_{\text{inter}}(\alpha) &= \langle \varphi X_{\text{inter}} | K^{\text{att}} | \varphi X_{\text{inter}} \rangle. \end{aligned} \quad (\text{H21})$$

These quantities contain the momentum dependence of the screened interaction, the interlayer separation factor, and the common envelope. Their average and difference are

$$\begin{aligned} \mathcal{V}(\alpha) &= \frac{\mathcal{V}_{\text{intra}}(\alpha) + \mathcal{V}_{\text{inter}}(\alpha)}{2}, \\ \Delta \mathcal{V}(\alpha) &= \frac{\mathcal{V}_{\text{intra}}(\alpha) - \mathcal{V}_{\text{inter}}(\alpha)}{2}. \end{aligned} \quad (\text{H22})$$

Within the two-channel equal-SRLP subspace, the attractive electron-hole interaction is

$$\begin{aligned} K_+^{\text{att}}(\alpha) &= \mathcal{V}_{\text{intra}}(\alpha) \mathcal{P}_{\text{intra}}^{\text{proj}} + \mathcal{V}_{\text{inter}}(\alpha) \mathcal{P}_{\text{inter}}^{\text{proj}} \\ &= \begin{pmatrix} \mathcal{V}(\alpha) & Q(\alpha) \Delta \mathcal{V}(\alpha) \\ Q(\alpha) \Delta \mathcal{V}(\alpha) & \mathcal{V}(\alpha) \end{pmatrix}. \end{aligned} \quad (\text{H23})$$

The average attraction $\mathcal{V}(\alpha)$ shifts the two excitons by the same amount. The difference between the intralayer and interlayer attraction energies couples them.

Because the attraction enters the BSE Hamiltonian with a minus sign, the coupling strength is

$$\Omega_+(\alpha) = Q(\alpha) |\Delta \mathcal{V}(\alpha)|. \quad (\text{H24})$$

In the AFM state,

$$\Omega_+ \equiv \Omega_+(0) = Q_0 \frac{|\mathcal{V}_{\text{intra}}(0) - \mathcal{V}_{\text{inter}}(0)|}{2}. \quad (\text{H25})$$

Equation (H25) assumes the same normalized relative-motion envelope for the two equal-SRLP excitons and uses the layer matrix elements of their conduction and valence states at the Γ point. If their relative-motion envelopes differ, the coupling must be obtained from the off-diagonal BSE matrix element.

4. Expansion of the reduced two-channel exciton model

Along the physical canting trajectory, we use

$$\mathcal{B} = \sin \alpha = \frac{B_{\text{ext}}}{B_{\text{sat}}} \in [0, 1]. \quad (\text{H26})$$

From Eq. (26),

$$\Delta_{\lambda\lambda'}(\mathcal{B}) = \Delta_{\Gamma}^0 - R_{A\lambda}(\mathcal{B}) - R_{B\lambda'}(\mathcal{B}). \quad (\text{H27})$$

The two equal-SRLP direct gaps satisfy

$$\begin{aligned} \Delta_{-,-}(\mathcal{B}) &= \Delta'_{\Gamma} - \chi_+ \mathcal{B} + O(\mathcal{B}^2), \\ \Delta_{+,+}(\mathcal{B}) &= \Delta'_{\Gamma} + \chi_+ \mathcal{B} + O(\mathcal{B}^2), \end{aligned} \quad (\text{H28})$$

where χ_+ is given in Eq. (54). Thus, to leading order in the single-particle expansion, the two equal-SRLP direct gaps acquire opposite linear canting shifts $-\chi_+ \mathcal{B}$ and $+\chi_+ \mathcal{B}$.

Choosing the relative phase of the two channel states so that the off-diagonal matrix element is real, the reduced two-channel Hamiltonian used in Eq. (52) is

$$H_+(\mathcal{B}) = (E_{0,X} + \xi_X \mathcal{B}^2) \mathbb{I}_2 + \begin{pmatrix} -\chi_+ \mathcal{B} & -\Omega_+ \\ -\Omega_+ & +\chi_+ \mathcal{B} \end{pmatrix}. \quad (\text{H29})$$

Here, $E_{0,X} + \xi_X \mathcal{B}^2$ is the average energy of the two equal-SRLP excitons when their mutual off-diagonal coupling is omitted, $2\chi_+ \mathcal{B}$ is their energy difference, and Ω_+ is the magnitude of the off-diagonal coupling between them.

The corresponding exciton energies are

$$E_{\pm,X}(\mathcal{B}) = E_{0,X} + \xi_X \mathcal{B}^2 \pm \sqrt{\Omega_+^2 + \chi_+^2 \mathcal{B}^2}. \quad (\text{H30})$$

For the reduced model, the expansion parameter of the square-root term is $|\chi_+ \mathcal{B} / \Omega_+|$. Therefore, when $\Omega_+ > |\chi_+|$, the expansion converges throughout the physical canting range $0 \leq \mathcal{B} \leq 1$. Retaining terms through second order gives

$$E_{-,X}(\mathcal{B}) = E_{0,X} - \Omega_+ + \left(\xi_X - \frac{\chi_+^2}{2\Omega_+} \right) \mathcal{B}^2 + O(\mathcal{B}^4). \quad (\text{H31})$$

The term $-\chi_+^2/(2\Omega_+)$ is the additional quadratic contribution generated by the coupling between the two equal-SRLP excitons.

5. Opposite-SRLP two-channel exciton model

The opposite-SRLP sector $\Lambda_{eh} = -1$ contains the two electron-hole channels $\{|c_+ v_- \rangle, |c_- v_+ \rangle\}$. Within the same common-envelope projection, the layer matrix elements connecting c_+ with c_- and v_+ with v_- have magnitudes $Q_A(0)$ and $Q_B(0)$, respectively. Their product is therefore the same band-edge factor $Q_0 = Q_A(0)Q_B(0)$ defined in Eq. (H16).

Using Eq. (30), we define

$$\chi_- = \frac{J^A t_{\perp}^A}{R_A} - \frac{J^B t_{\perp}^B}{R_B}. \quad (\text{H32})$$

The two opposite-SRLP direct gaps then have opposite leading canting shifts,

$$\begin{aligned} \Delta_{+,-}(\mathcal{B}) &= \Delta'_{\Gamma} - \chi_- \mathcal{B} + O(\mathcal{B}^2), \\ \Delta_{-,+}(\mathcal{B}) &= \Delta'_{\Gamma} + \chi_- \mathcal{B} + O(\mathcal{B}^2). \end{aligned} \quad (\text{H33})$$

Let $\mathcal{V}_{\text{intra}}^-$ and $\mathcal{V}_{\text{inter}}^-$ denote the intralayer and interlayer electron-hole attraction energies obtained using the common relative-motion envelope for these two channels. Their difference couples the two opposite-SRLP excitons. The reduced two-channel Hamiltonian is

$$H_{\text{eff}}^-(\mathcal{B}) = (E_{0,X}^- + \xi_X^- \mathcal{B}^2) \mathbb{I}_2 + \begin{pmatrix} -\chi_- \mathcal{B} & -\Omega_- \\ -\Omega_- & +\chi_- \mathcal{B} \end{pmatrix}, \quad (\text{H34})$$

$$\Omega_- = Q_0 \frac{|\mathcal{V}_{\text{intra}}^- - \mathcal{V}_{\text{inter}}^-|}{2}.$$

Here, $E_{0,X}^-$ is the common energy of the two opposite-SRLP excitons at the AFM state when their mutual coupling is omitted, $\xi_X^- \mathcal{B}^2$ is the quadratic energy shift common to both excitons, and Ω_- is the magnitude of their off-diagonal coupling.

The corresponding exciton energies are

$$E_{\pm,X}^-(\mathcal{B}) = E_{0,X}^- + \xi_X^- \mathcal{B}^2 \pm \sqrt{\Omega_-^2 + \chi_-^2 \mathcal{B}^2}. \quad (\text{H35})$$

The expansion parameter of the square-root term is $|\chi_- \mathcal{B} / \Omega_-|$. If $\Omega_- > |\chi_-|$, the expansion converges throughout the physical canting range $0 \leq \mathcal{B} \leq 1$. Retaining terms through second order for the lower exciton gives

$$E_{-,X}^-(\mathcal{B}) = E_{0,X}^- - \Omega_- + \left(\xi_X^- - \frac{\chi_-^2}{2\Omega_-} \right) \mathcal{B}^2 + O(\mathcal{B}^4). \quad (\text{H36})$$

The term $-\chi_-^2/(2\Omega_-)$ is the additional quadratic contribution generated by the coupling between the two opposite-SRLP excitons.

At the AFM state $\mathcal{B} = 0$, the two opposite-SRLP excitons are degenerate when their mutual coupling is omitted. Including the coupling gives the energies $E_{0,X}^- - \Omega_-$ and $E_{0,X}^- + \Omega_-$, separated by $2\Omega_-$.

The opposite-SRLP curves in Fig. 7(a) are obtained from Eq. (H34). Within the minimal CrSBr Hamiltonian, these excitons remain optically dark because the conduction and valence states forming each channel have opposite SRLP eigenvalues.

6. Layer probabilities of the equal- and opposite-SRLP excitons

The physical intralayer and interlayer probabilities are the expectation values of the two-particle layer operators

introduced in Eq. (58). We evaluate these probabilities directly for the lower-energy eigenstate of each reduced two-channel Hamiltonian.

For the equal-SRLP sector, we use the ordered basis $\{|c_-v_- \rangle, |c_+v_+ \rangle\}$ and write the lower-energy exciton as

$$|X_{+,low}(\mathcal{B})\rangle = \mathbf{a}_+(\mathcal{B})|c_-v_- \rangle + \mathbf{b}_+(\mathcal{B})|c_+v_+ \rangle. \quad (\text{H37})$$

Diagonalizing Eq. (H29) gives

$$\begin{aligned} \mathbf{a}_+^2(\mathcal{B}) &= \frac{1}{2} \left[1 + \frac{\chi_+\mathcal{B}}{\sqrt{\Omega_+^2 + \chi_+^2\mathcal{B}^2}} \right], \\ \mathbf{b}_+^2(\mathcal{B}) &= \frac{1}{2} \left[1 - \frac{\chi_+\mathcal{B}}{\sqrt{\Omega_+^2 + \chi_+^2\mathcal{B}^2}} \right], \\ 2\mathbf{a}_+(\mathcal{B})\mathbf{b}_+(\mathcal{B}) &= \frac{\Omega_+}{\sqrt{\Omega_+^2 + \chi_+^2\mathcal{B}^2}}. \end{aligned} \quad (\text{H38})$$

Equation (H18) then gives the interlayer probability

$$w_{inter,+}^{low}(\mathcal{B}) = \frac{1}{2} \left[1 - Q_0 \frac{\Omega_+}{\sqrt{\Omega_+^2 + \chi_+^2\mathcal{B}^2}} \right]. \quad (\text{H39})$$

The corresponding intralayer probability is

$$w_{intra,+}^{low}(\mathcal{B}) = 1 - w_{inter,+}^{low}(\mathcal{B}). \quad (\text{H40})$$

This is the expression used in Eq. (60). When $\Omega_+ > |\chi_+|$, its expansion converges throughout the physical interval $0 \leq \mathcal{B} \leq 1$ and gives

$$w_{inter,+}^{low}(\mathcal{B}) = \frac{1 - Q_0}{2} + \frac{Q_0\chi_+^2}{4\Omega_+^2}\mathcal{B}^2 + O(\mathcal{B}^4). \quad (\text{H41})$$

For the opposite-SRLP sector, we use the ordered basis $\{|c_+v_- \rangle, |c_-v_+ \rangle\}$ and write

$$|X_{-,low}(\mathcal{B})\rangle = \mathbf{a}_-(\mathcal{B})|c_+v_- \rangle + \mathbf{b}_-(\mathcal{B})|c_-v_+ \rangle. \quad (\text{H42})$$

Diagonalizing Eq. (H34) gives

$$\begin{aligned} \mathbf{a}_-^2(\mathcal{B}) &= \frac{1}{2} \left[1 + \frac{\chi_-\mathcal{B}}{\sqrt{\Omega_-^2 + \chi_-^2\mathcal{B}^2}} \right], \\ \mathbf{b}_-^2(\mathcal{B}) &= \frac{1}{2} \left[1 - \frac{\chi_-\mathcal{B}}{\sqrt{\Omega_-^2 + \chi_-^2\mathcal{B}^2}} \right], \\ 2\mathbf{a}_-(\mathcal{B})\mathbf{b}_-(\mathcal{B}) &= \frac{\Omega_-}{\sqrt{\Omega_-^2 + \chi_-^2\mathcal{B}^2}}. \end{aligned} \quad (\text{H43})$$

The off-diagonal layer matrix element in this basis contains the same product $Q_A(0)Q_B(0) = Q_0$ as in the equal-SRLP sector. The interlayer probability of the lower opposite-SRLP exciton is therefore

$$w_{inter,-}^{low}(\mathcal{B}) = \frac{1}{2} \left[1 - Q_0 \frac{\Omega_-}{\sqrt{\Omega_-^2 + \chi_-^2\mathcal{B}^2}} \right]. \quad (\text{H44})$$

The corresponding intralayer probability is

$$w_{intra,-}^{low}(\mathcal{B}) = 1 - w_{inter,-}^{low}(\mathcal{B}). \quad (\text{H45})$$

Expanding about the AFM state gives

$$w_{inter,-}^{low}(\mathcal{B}) = \frac{1 - Q_0}{2} + \frac{Q_0\chi_-^2}{4\Omega_-^2}\mathcal{B}^2 + O(\mathcal{B}^4). \quad (\text{H46})$$

This expansion converges when $|\chi_-\mathcal{B}/\Omega_-| < 1$. It therefore applies throughout $0 \leq \mathcal{B} \leq 1$ only when $\Omega_- > |\chi_-|$. The exact expression in Eq. (H44) does not require this condition.

Equations (H39) and (H44) give the layer probabilities plotted for the lower equal- and opposite-SRLP excitons in Fig. 7(b).

7. Electron-hole continuum edge and exciton binding energy

The direct BSE Hamiltonian can be written as

$$H_{\text{BSE}} = H_0^{eh} - K^{\text{att}}, \quad (\text{H47})$$

where H_0^{eh} contains the independent electron-hole energies and K^{att} denotes the attractive direct electron-hole interaction.

For a normalized eigenstate,

$$E_{n,X} = \langle \Psi_n | H_0^{eh} | \Psi_n \rangle - \langle \Psi_n | K^{\text{att}} | \Psi_n \rangle. \quad (\text{H48})$$

Therefore,

$$\langle \Psi_n | H_0^{eh} | \Psi_n \rangle - E_{n,X} = \langle \Psi_n | K^{\text{att}} | \Psi_n \rangle. \quad (\text{H49})$$

The right-hand side is the mean attractive interaction energy of the exciton. It is not its binding energy, because the electron-hole wave function also has an independent-particle energy above the bottom of the continuum.

For each SRLP sector $\Lambda_{eh} = \pm 1$, the continuum edge is the lowest energy of an independent electron-hole pair in that sector,

$$E_{\text{th}}^{(\Lambda_{eh})}(\mathcal{B}) = \min_{\mathbf{k}, (c,v) \in \mathcal{C}_{\Lambda_{eh}}} [E_c(\mathbf{k}, \mathcal{B}) - E_v(\mathbf{k}, \mathcal{B})]. \quad (\text{H50})$$

An exciton is bound when $E_{n,X}^{(\Lambda_{eh})}(\mathcal{B}) < E_{\text{th}}^{(\Lambda_{eh})}(\mathcal{B})$. Its binding energy is

$$E_{b,n}^{(\Lambda_{eh})}(\mathcal{B}) = E_{\text{th}}^{(\Lambda_{eh})}(\mathcal{B}) - E_{n,X}^{(\Lambda_{eh})}(\mathcal{B}). \quad (\text{H51})$$

Combining Eqs. (H48) and (H51) gives

$$E_{b,n}^{(\Lambda_{eh})} = \langle \Psi_n | K^{\text{att}} | \Psi_n \rangle - \left[\langle \Psi_n | H_0^{eh} | \Psi_n \rangle - E_{\text{th}}^{(\Lambda_{eh})} \right]. \quad (\text{H52})$$

The binding energy is therefore the attractive interaction energy reduced by the independent electron-hole energy required to localize the relative motion above the continuum edge.

8. Validity of the two-channel exciton model

The separation of the BSE into equal- and opposite-SRLP sectors remains exact for the layer-symmetric direct electron-hole interaction. The reduced two-channel exciton models require three additional conditions.

First, the layer composition of the conduction and valence states must vary weakly over the momenta that contribute appreciably to the exciton.

For $\xi = c, v$, we write

$$U_c(\mathbf{k}) = (|c_-(\mathbf{k})\rangle, |c_+(\mathbf{k})\rangle). \quad (\text{H53})$$

$$U_v(\mathbf{k}) = (|v_-(\mathbf{k})\rangle, |v_+(\mathbf{k})\rangle). \quad (\text{H54})$$

$$\mathcal{L}_\ell^{(\xi)}(\mathbf{k}) = U_\xi^\dagger(\mathbf{k}) \mathbb{P}_\ell U_\xi(\mathbf{k}). \quad (\text{H55})$$

The band-edge layer matrices are representative of the states sampled by the exciton when

$$\max_{\substack{\mathbf{k} \in \text{supp } \varphi \\ \ell=1,2; \xi=c,v}} \left\| \mathcal{L}_\ell^{(\xi)}(\mathbf{k}) - \mathcal{L}_\ell^{(\xi)}(\Gamma) \right\| \ll 1. \quad (\text{H56})$$

Under this condition, the layer matrix elements evaluated at Γ can be used over the momentum range occupied by the exciton wave function.

Second, electron-hole states involving c'_λ or v'_λ must remain sufficiently separated in energy from the four near-gap channels retained in the BSE. For the two reduced

SRLP models, a sufficient condition is

$$\Delta_{\text{remote}} \gg \max(E_b, \Omega_+, \Omega_-, |\chi_+ \mathcal{B}|, |\chi_- \mathcal{B}|). \quad (\text{H57})$$

For the CrSBr parameters, the smaller band-edge separation to a remote single-particle band is $2R_A = 0.523$ eV, compared with an equal-SRLP exciton binding energy near 0.197 eV and an equal-SRLP coupling Ω_+ near 0.051 eV.

The separation is therefore much larger than the equal-SRLP coupling but only a few times larger than the binding energy. The accuracy of the two-channel reduction is consequently assessed directly by comparison with the full BSE in Fig. 7.

Third, the two excitons retained within each SRLP sector must have similar relative-motion wave functions. For the equal-SRLP sector, this requires

$$|\langle \varphi_-, - | \varphi_+, + \rangle| \simeq 1. \quad (\text{H58})$$

For the opposite-SRLP sector, the corresponding condition is

$$|\langle \varphi_+, - | \varphi_-, + \rangle| \simeq 1. \quad (\text{H59})$$

These conditions allow the two excitons within each SRLP sector to share a common relative-motion envelope while retaining their different band and layer-spin compositions.

If these conditions are not satisfied, the equal- and opposite-SRLP sectors remain separated by symmetry, but the two-channel Hamiltonians need not reproduce the exciton energies or layer probabilities quantitatively. The couplings and the intralayer and interlayer probabilities must then be obtained directly from the full momentum-dependent BSE eigenstates.

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